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Integration and substitution in hybrid manufacturing and refurbishing systems

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ABSTRACT

Inspired by a real-work case, this paper introduces a profit maximization model for dynamic lot sizing considering substitution and multiple usage of returns for refurbishing at different levels of quality or for disassembly to extract key parts that will be used in the manufacturing process. This model allows studying the interactions between different types of returns and decisions. To analyze the impact of uncertainty on these interactions, we develop a robust two-stage stochastic program with uncertainty on demands and returns. The resulting problems are mixed-integer linear programs that we solve using an efficient relax-and-fix and fix-and-optimize heuristic. Extensive numerical experiments are conducted to study the different trade-offs when integrating multi-usage of returns with substitution and accordingly derive managerial insights. The experiments have revealed, for example, that: (i) the profit margin of the refurbished items is the main determinant of the total profit and when such margin in high, the total profit becomes more sensitive to different cost variations; (ii) collection efficiency becomes very sensitive to collection cost and much less sensitive to refurbishing cost especially with large profit margins; (iii) when demand and returns are uncertain parameters, substitution becomes the best option as uncertainty and prices increase; (iv) when the gap between prices of different quality levels is high, lost sales occur mainly on the lowest quality product and downgrading decreases together with substitution; and (v) Sharing the production line between refurbishing and manufacturing for low-quality products is highly motivated by small upgrading costs and their substitution level increases with increasing upgrading costs.

1. Introduction

In 2022, there was 62 billion kg of e-waste (electronic waste) worldwide. This quantity can cover the area of an entire city.[1] These wastes can be considerably reduced if circular economy principles are implemented by companies, moving from the traditional linear systems (take, make, dispose) to a closed loop system in which products are reused, shared, repaired, refurbished, remanufactured and recycled (Invernizzi et al., 2020). Companies are shifting to circular systems because of pressures from the society and governments in addition to the scarcity of resources. Hence companies have been urged to accelerate their search for more sustainable solutions (Rogers et al., 1999; Krajewski et al., 2016). This is the case for a large consumer electronics with which the authors of this paper have collaborated. The company was looking for solutions to develop its closed-loop supply chain with the most efficient reverse logistics processes. The authors collaborated with the company to develop mathematical models and efficient solution procedures for emerging optimization problems. In addition to the aforementioned reasons for implementing circularity, this company was motivated by the utilization of its resources. It has made large investments in acquiring new assembly lines that will become largely under utilized in the near future because of the shortage of imported parts.

Hence, the purpose of this collaboration between the authors and the company is to help the latter in the design of a closed-loop supply chain by modeling and solving different but interrelated logistics and production planning problems. We started with the initial idea of the company which consists of using one line for manufacturing new products and the newly installed line for refurbishing returned products. This model was then extended by proposing *downward substitution* at the finished items level. By downward substitution we mean that the

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[1] https://www.itu.int/en/ITU-D/Environment/Pages/Publications/The-Global-E-waste-Monitor-2024.aspx.

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demand for refurbished products can be satisfied using new items. The downward substitution at the finished goods level occurs in practice when there are different market segments for new and refurbished products. In this situation a customer's demand for some refurbished product arrives when such product is out of stock. The planner may choose to use the inventory of some new product to fulfill the demand and charge the price of the refurbished product. This can avoid shortage and setup cost of directly producing the refurbished product. It can also lead to opportunities to create greater flexibility in their production and inventory planning.

The objective is to maximize profit while fully satisfying the demand for new products and attempting to meet the demand for used (refurbished) products. The profit is calculated as the difference between the sales revenue and the total costs, which include disassembly, production and inventory holding costs among others. The demand for new products must be fully satisfied while shortages are allowed for used products. Over a discrete time horizon with a dynamic demand, the planner must decide for each time period the quantity of items that can be collected and stocked to enter the process, the quantity to be disassembled and whose key component is used for manufacturing new products, the quantity of the new key part to be purchased for the manufacturing of new products, the quantity of used items that will be refurbished, and the quantity of new items that will be used for substitution. From the disassembly process, we focus on the extraction of a single key part, resulting in a 1:1:1 relationship, i.e. one unit of a disassembled returned item generates one key part, which will be used in the manufacturing of one unit of the finished new item. This study explicitly distinguishes between refurbishing (used second hand finished products) and remanufacturing (spare parts) following the framework and terminology proposed by Thierry et al. (1995). The latter proposed that return options can be divided into repair, refurbishing, remanufacturing, cannibalization, and recycling. Using old parts in the assembly of new products is considered as cannibalization and remanufacturing, according to the definitions presented by Thierry et al. (1995). More details on the description and formulation of the problem are presented in Section 3.1.

This study can be divided into three main parts: A first part where a single type of products is sold either as a used refurbished item or as a new item. In the second part, a stochastic model with uncertainty on the collected items and demand for used items is proposed. A two-stage stochastic model is used in this case and several managerial insights are provided. In this same part, a risk averse stochastic programming model is proposed. Finally, we discuss different trade-offs that take place when products of different values and quality grading are considered in the process. Such a system results in more complex flows between different processes in the system. The paper also presents an efficient heuristic to solve the corresponding planning problems and analyzes the economic and ecological advantages of substitution and integration of decisions between used and new products.

The main contributions of the paper can be summarized as follows:

- It provides a literature review specific to lot sizing with refurbishing and substitution, which has not been done before, to the best of our knowledge.
- It introduces a profit maximization integrated model for dynamic lot sizing considering substitution and multiple usage of returns for refurbishing at different levels of quality (through upgrading or downgrading) or for disassembly to extract key parts that will be used in the manufacturing process. The model was inspired by a real case from industry.
- It proposes a stochastic optimization model with uncertainty on demand for used items and on the returned items.
- It proposes an efficient relax-and-fix and fix-and-optimize heuristic to solve large instances of the integrated model.
- It presents extensive numerical analysis and derives managerial insights that show the importance of substitution, multi-usage of returns, and consideration of uncertainty and multi-grades of products.

Despite the increase in research on reverse logistics, to the best of our knowledge, there are no publications on the integrated decisions of the production planning of new products, refurbished products (with lower value than the new ones), and parts (as good as new parts) obtained from disassembly of returned products, alongside downward substitution (demand for refurbished products satisfied using new products). However, as indicated above, companies are interested in using existing assembly lines for disassembly and refurbishing purposes, justifying the integration of such decisions. Furthermore, downward substitution is a significant tool used in the semiconductor industry and computer retail (e.g. Ervolina et al., 2006; Gallego et al., 2006).

The remainder of the paper is organized as follows. Section 2 presents a review of recent publications relevant to our work. The problem is described in detail and formulated in Section 3. Section 4 details the heuristic solution approach. Section 5 presents the numerical experiments used to analyze the performance of the heuristic approach. This section also presents several managerial insights. Finally, conclusions are presented in Section 6.

2. Literature review

This section summarizes the publications relevant to our study. We start by presenting general papers related to reverse logistics and hybrid systems with substitution. Then, we present some production planning problems with returns, and we focus on studies that address the lot sizing problem with returns with substitution. The main aspects of these studies are summarized in Table 1.

2.1. Reverse logistics and hybrid production systems with substitution

Recent literature surveys on reverse logistics and closed loop supply chains have shown that thousands of scientific articles have been published on these topics during the last 30 years. Some of these surveys include those of Guide and Van Wassenhove (2009), Govindan et al. (2015), Govindan and Soleimani (2017), and Kazemi et al. (2019). For example, Rizova et al. (2020) identified hundreds of relevant publications in recent years. This literature review focuses on studies related to our research, namely production planning decisions related to returns rather than the design of closed loop supply chains and planning in reverse logistics in general.

The literature on the processing options for returned products presents different streams and terminologies. For example, although there is a difference between the terms "refurbishing" (or "refurbishment") and "remanufacturing", several authors have been using them interchangeably (e.g. Vorasayan and Ryan, 2006). It can be deduced that there are two main streams. The first follows the classification and terminology proposed by Thierry et al. (1995), which has been considered by several authors (e.g., Gungor and Gupta, 1999; King et al., 2006; Zikopoulos and Tagaras, 2007; Konstantaras et al., 2010; Rachih et al., 2019). The article by Zacharaki et al. (2020), is related to a recent project funded by the European Commission. Thierry et al. (1995) proposed that product recovery can take the following forms: repair, refurbishing, remanufacturing, cannibalization, and recycling. In the second stream, researchers follow a simple categorization initially proposed by Fleischmann et al. (1997), which classifies recovery into recycling (material recovery) and remanufacturing (value-added recovery). See also Guide and Van Wassenhove (2009).

We followed the terminology used by Thierry et al. (1995). In particular, distinction is clearly made between *refurbishing* where critical modules are inspected and upgraded to a specified quality level, and *Remanufacturing* where modules and parts are inspected and upgraded to as good as new quality.

Companies may use their new product manufacturing plants to disassemble, remanufacture, or refurbish used products. These systems are referred to as hybrid systems. They can take different forms. For

Table 1
Summary of literature.

Author	Dem	Plan. Hor.	Ret inv	Comp. inv.	Proc. Capa.	Setups	Activity type[a]	Usages of returns	Subs.	Objectives	Solution approach
Inderfurth (2004)	sto	Single period	Yes	No	No	No setup	mf, rmf	Mono	Down	Max Profit	Markov
Li et al. (2006)	det	Disc, fin	Yes	No	No	Separate	mf, rmf	Mono	Down	Min Cost	DP
Li et al. (2007)	det	Disc, fin	Yes	No	Yes	Separate	mf, rmf	Mono	Down	Min Cost	GA, DP
Bayındır et al. (2007)	sto	Single period	No	No	Yes, Shared	No setup	mf, rmf	Mono	Down	Max Profit	Markov
Piñeyro and Viera (2010)	det	Disc, fin	No	No	No	Separate	mf, rmf	Mono	Down	Min Cost	Tabu Search
Marshall and Archibald (2015)	sto	Cont., inf	Yes	Yes	Yes	Separate	mf, rmf, cann	Mono	Bidirec	Max Profit	Semi-Markov Decision Process
Ahiska et al. (2017)	sto	Cont., inf	Yes	No	Yes	Separate	mf, rmf	Mono	Down	Max Profit	Heuristic
Fang et al. (2017)	sto	Cont., inf	Yes	Yes	Yes	No setup	mf, rmf, cann	Mono	Down	Max Profit	Multi-variate optim.
Koken et al. (2018)	det	Disc, fin	Yes	No	Yes	Separate	mf, rmf	Mono	Down	Max Profit	GA, Others
Liu et al. (2019)	sto	Disc, fin	Yes	No	Yes	Separate	mf, rmf, comp	Mono	Down	Min Cost	Ant Colony Optimization
Afshar-Bakeshloo et al. (2021)	sto	Cont., inf.	No	No	Yes	No setup	mf, rmf1 and rmf2	Mono	Down (3 levels)	Max Profit, customer satisf.	Markov
Liu and Papier (2022)	sto	Cont, inf	Yes	No	Yes	No setup	mf, rmf	Mono	Bidirec	Max Profit	Markov Decision process
Our work	det, sto	Disc, fin	Yes	Yes	Yes, Shared	Joint	mf, rfb, cann, comp	Multi	Down	Max Profit	Relax-and-fix and fix-and-optimize

[a] mf: manufacturing, rmf: remanufacturing, cann: cannibalization, comp: component replenishment.

example, the same assembly line of new products can be used to disassemble, refurbish, or remanufacture returned products. In this case, the changeover times and costs should be considered. Other systems may use separate lines for new and used products.

Managing production in hybrid systems is more challenging than in classical assembly lines of new products. Difficulties can arise from the complexity of the design decisions (e.g., Gungor and Gupta, 1998) and planning decisions on which items to refurbish or remanufacture and which parts to disassemble or recycle (Teunter et al., 2006; Cunha and Melo, 2016). Another source of complexity is the uncertainty in both the demand and returns of the used products (Fang et al., 2017; Naeem et al., 2013; Assid et al., 2019; Macedo et al., 2016; Kilic and Tunc, 2019), the uncertainly in refurbishing duration (Slama et al., 2022), and the uncertainty on the quality of returned products (Quezada et al., 2020). Furthermore, such systems become harder to model and solve when considering different types of products with different levels of quality and when substitution is allowed between products.

Substitution, particularly downward substitution, has been used as a strategy in the automotive industry (Ahiska and Kurtul, 2014). Earlier studies on substitution that did not consider returns or reverse logistics include Mcgillivray and Silver (1978), Pasternack and Drezner (1991) and Benkherouf et al. (2017). In dynamic demand models, contributions include those of Balakrishnan and Geunes (2000), Geunes (2003), Yaman (2009) and Jing and Mu (2020), in addition to the book by Lang and Domschke (2010). Shin et al. (2015) present a recent survey and literature classification. Substitution can occur both at the early stages of the production process (choice between used and new parts) and at the finished goods level. Substitution at the early stages raises several legal and technical issues. If it is possible to choose between a new purchased part and a used part which has been inspected and is considered as good as new (this is case in our paper), we would not consider this as substitution as it will not be possible to distinguish between finished goods containing either type of parts. If the parts are just refurbished (have undergone less rigorous inspections compared to remanufactured part), then the finished product cannot be considered as new. This is different from our model. As indicated in the Introduction section of the paper, downward substitution at the finished goods level occurs when there are different market segments for new and refurbished products. Industrial applications where segmented market for manufactured new products and refurbished products occurs include photocopiers, tires and personal computers (Piñeyro and Viera, 2010; Li et al., 2006) and memory chips (Leachman and Glassey, 1987) It should be noted that despite the large number of publications on production planning for remanufacturing and hybrid systems, few publications consider substitution.

2.2. Dynamic lot sizing with product returns

The problem under consideration in this study is directly related to the lot sizing problem with product returns (Teunter, 2004) and more particularly to the dynamic version of the problem (Teunter et al., 2006). The earliest known publication on the dynamic lot sizing problem is the seminal paper by Wagner and Whitin (1958). They proposed dynamic programming algorithms for the single item lot sizing problem with time varying demand and without capacity restrictions. Several extensions of this problem have been reported in the literature. Researchers have considered problems such as limited production capacity, parameter uncertainty, non-linear costs, and lost sales. Recent comprehensive surveys have been conducted on lot sizing problems and their extensions, complexity, and solution approaches (Jans and Degraeve, 2008; Buschkühl et al., 2010; Brahimi et al., 2017; Gicquel et al., 2008; Quadt and Kuhn, 2008; Suwondo and Yuliando, 2012). Most of the extensions of the uncapacitated single item lot sizing problem are NP-hard, although polynomial algorithms have been developed for certain special cases. For example, the single item capacitated dynamic lot sizing problem is NP-hard in general, but it becomes polynomial when the capacity is constant over all time periods. The consideration of lot sizing with returns is one of the most important research directions recommended in the survey by Brahimi et al. (2017). Here, a review of publications in this area is given with focus on the most relevant to current paper.

When considering problems with returns, the corresponding publications can be broadly classified into discrete and continuous models.

The discrete models that address the dynamic lot sizing problem with product returns were first developed by Richter and Sombrutzki (2000), Richter and Weber (2001), and Teunter et al. (2006). Inventory control models with returns were discussed by Inderfurth (2004), Jaber and El Saadany (2009), Mitra (2016), Wang et al. (2017), and Fang et al. (2017).

Because of the complexity of lot sizing decisions in remanufacturing/refurbishing and hybrid systems, most proposed solution approaches in the literature are heuristics. An exception is the polynomial time dynamic programming algorithm proposed by Teunter et al. (2006), for a problem with separate setups and time invariant costs, and Arulselvan and van den Heuvel (2022) for a problem with certain realistic assumptions on the cost structure. Retel Helmrich et al. (2014) discussed the complexity of dynamic lot sizing problems with returns and showed that all lot sizing problems with returns are NP-hard, except for the problem solved by Teunter et al. (2006). Heuristics used to solve NP-hard versions of this problem include extensions of the classical lot sizing heuristics such as Silver and Meal (1973) and Schulz (2011); a dynamic programming based heuristic proposed by Fazle Baki et al. (2014); metaheuristics such as the variable neighborhood search proposed by Sifaleras et al. (2015), the tabu search heuristic developed by Piñeyro and Viera (2009); a memetic algorithm and a hybrid algorithm suggested by Zouadi et al. (2015). For the multi-item problem, Sahling (2013) developed a column generation approach with particular capacity constraints, whereas (Sifaleras and Konstantaras, 2017) solved another variant of the multi-item lot sizing problem with returns using a variable neighborhood descent heuristic.

We will now discuss previous research work on lot sizing with returns and substitution. Though our focus will be on dynamic lot sizing models, we will present contributions on inventory control models considering returns and substitution. The closest references to our study are summarized in Table 1. The table lists the main attributes of each paper. References are classified as follows:

- Demand (dem): stochastic (sto) or deterministic (det).
- The planning horizon (horizon): finite (fin) or infinite (inf), discrete (disc), continuous (cont) or single period.
- With returns inventory (ret inv): Yes/No.
- Component inventory (Comp. inv.): Yes/No.
- Processing Capacities (proc capa): “No” if the capacity is infinite, “Yes” if any of the processes is capacitated, “Shared” if several processes share the same resource(s).
- Setups: indicate cases with “no setup” costs or times and cases with “separate” or “joint” setups;
- Activity type: manufacturing (mf), remanufacturing (rmf), refurbishing (rfb), cannibalization (cann), and new component replenishment (comp).
- Usage of Returns indicates whether returns are used in both manufacturing and remanufacturing/refurbishing processes (Multi) or not (Mono).
- Substitution: whether it is downward, upward, or bidirectional substitution.
- Objectives: indicates number and type of objectives.
- Solution approaches: indicates which solution is used to solve the problem.

Table 1 shows that only six (out of 12 references) have proposed dynamic lot sizing models with returns and substitution. Also, more than half of the models have stochastic demands (8 out of 12 references). All these references consider continuous time or single period time horizons except for the study by Liu et al. (2019). The latter assume that both the amount of returns and demand for remanufactured items are stochastic. They proposed an ant colony algorithm with a random sampling method to solve the problem. Despite the importance and challenges of storing returns, some studies have ignored these aspects. For example, Bayındır et al. (2007) assumed that there is a continuous flow of returns with a fixed cost that can be included directly in the remanufacturing cost.

To the best of our knowledge, only two papers consider cannibalization, i.e. the use of extracted parts from a returned items through disassembly: Fang et al. (2017) and Marshall and Archibald (2015). Compared to our work, the models in these two references assume infinite and continuous time horizons. In the current work, we study a dynamic demand finite time horizon model which is more suitable for production planning with high demand variability. While Inderfurth (2004), Li et al. (2006), and Piñeyro and Viera (2010) assumed infinite processing capacity, all the other references in Table 1 assume finite capacity processes. In all these studies, different processes (e.g., manufacturing and remanufacturing) have independent resources, each of which has its own capacity.

In our contribution, we compare models with finite and infinite capacities and analyze the impact of line sharing on the decisions. With respect to setup costs and time, there are no studies on joint setups between manufacturing and remanufacturing/refurbishing even in the case of shared capacity presented by Bayındır et al. (2005) in their single period model. Further, we analyzed models with joint and separate setup times and costs. Besides the fact that this is different from the initial idea of the company (having two completely independent lines), sharing the same production line for manufacturing and remanufacturing has shown its advantages in several situations (Tang and Teunter, 2006).

All studies on dynamic lot sizing models in Table 1 allow downward substitution, where the shortage of refurbished products is compensated by the higher value new product, but at the same price as the used one. Two inventory control models were proposed with bidirectional substitution. A particular case (also in a continuous time model) is that of Afshar-Bakeshloo et al. (2021) where the product (tires in this case) can have three different forms: new, remanufactured once, or remanufactured twice. The products are destined to prime, budget, and low budget market segments. The authors propose the successive downward substitution of products. Afshar-Bakeshloo et al. (2021) use Markov chains to the model the system. In the case of our study, we propose different levels of quality and allow *downgrading* and *upgrading* of returned products. In the refurbishing process, a premium returned electronic product, for example, can be downgraded by using lower quality components into it.

In terms of objectives, the references in Table 1 can be classified as profit maximization (e.g., Ahiska et al., 2017; Bayındır et al., 2005) or cost minimization (e.g., Li et al., 2006; Piñeyro and Viera, 2009). The results from (Li et al., 2006) show that the cost savings due to downward substitution can reach 15%. While continuous time models mostly use Markovian decision processes, dynamic lot sizing model use metaheuristics such as genetic algorithms (Li et al., 2007), tabu search (Piñeyro and Viera, 2010) and ant colony optimization (Liu et al., 2019). In our case, we have proposed an efficient relax-and-fix and fix-and-optimize heuristic.

The choice of the relax-and-fix and fix-and-optimize heuristic is motivated by its successful use in solving several lot sizing problems. Pochet and Wolsey (2006) presents a general scheme of a relax-and-fix heuristic applied to a generic mixed integer program. For lot sizing problems, relax-and-fix heuristics have been used by Absi and van den Heuvel (2019) for the uncapacitated single-item lot sizing problem, by Federgruen et al. (2007) for the capacitated single level multi-item lot sizing problem and by Stadtler (2003) and Akartunalı and Miller (2009) for a multi-level lot sizing problem. Fix-and-optimize heuristics have been applied by Sahling et al. (2009) and Helber and Sahling (2010) to a multi-level capacitated lot sizing problem. A combined approach based on fix-and-relax and fix-and-optimize was applied by Stadtler and Sahling (2013) to a lot sizing and scheduling model for multi-stage flow lines with zero lead times and by Toledo et al. (2015) to multi-level lot sizing problems. Further details regarding the heuristics are presented in Section 4.

2.3. Summary of contributions

Based on the above literature review and on Table 1, we can summarize the contributions and particular aspects of our work as follows. To the best of our knowledge, this is the first time that a hybrid system combining refurbishing, manufacturing, and disassembly processes is modeled and solved. We have justified the usefulness of such a model in practice through examples from industry.

Our paper is also the first to analyze a model with shared setups between the processing lines. Like several other studies, we have adopted downward substitution and we have shown the relevance of this choice in the previous paragraphs. This first deterministic model is followed by models where the amount of returns and refurbished item's demand are input stochastic parameters. A two-stage and risk averse two-stage models are presented and solved. Finally, a multi-item model with different quality and finishing levels is considered. The amount of returns is an input parameter and it is correlated with the demand for new items of the same grade. Downgrading and upgrading are allowed at different costs. A detailed discussion of flows between processes is provided in the numerical experiments section. To solve hard instances of the problems, an efficient mathematical programming-based heuristic is proposed. The heuristic can obtain solutions close to optimum in less than 5 s for reasonable size instances.

3. Problem description

Here we describe the problem illustrated in Fig. 1. In this hybrid system, the production facility receives returned products. One unit of the returned item is composed of one core part (component) and several non-core parts (see Liu et al., 2019). The core part can be regarded as good as new (by undergoing some processes). While the non-core parts cannot be regarded as good as new. To produce one unit of a refurbished item (Path 1), we use one unit of a returned item (core part and non-core parts). But if we want to use a returned item in manufacturing of new item (Path 2), we disassemble the non-core parts (waste recovery) from the returned item and keep the one core part to be used in the assembly of the new item. Some of the returns will be discarded and sent directly to an independent recycling process (not shown here). The accepted items are eventually stored in inventory and separated into two sets. One set of items is refurbished (Path 1 on Fig. 1) and used to satisfy the demand for second-hand products.

The second set of returned items are disassembled to extract the core part (key component) that can be regarded as good as new as indicated previously. Most prior studies implicitly assume that all returns are of the same condition, i.e. homogeneous quality, and thus require the same processing and holding costs. For example, from each unit of TVs returned, the part or subassemblies recovered may be: *control board*, *power cables*, *mother board*, etc. The key parts extracted from each unit of item returned are approximately the same. The group of components recovered from one unit of a returned item is called a *part* in the rest of paper. Each part is used in the manufacturing of one unit of the new product (Path 2). The latter can also be manufactured using the new purchased parts (Path 3). The remaining components and parts resulting from the disassembly can enter the recycling process. Given certain conditions, the company is also ready to substitute secondhand products with new ones at the same price, if this is economically feasible. Fig. 2 illustrates the one-to-one relationship between refurbished, disassembled, and manufactured products and parts. The upper part of the figure shows that for one unit of the returned items, the process generates one unit of refurbished items after fixing, replacing, and upgrading operations. The lower part shows that from one unit of the returned and disassembled item, we extract one unit of the "key" part A. The dashed lines indicate the alternative option of the planner to use this extracted part or a purchased one from the supplier.

Continuing with the description of the problem in Fig. 1, Path 3 indicates the case where the core part of the product is purchased instead of being recovered from a collected item. Finally, Path 4 represents the case where the refurbishing task is done on the same production line of the new items, requiring a changeover time between refurbishing and assembly of new items on this line.

The objective is to maximize the total profit calculated as the difference between the revenue generated from the sales of items (new and refurbished) and the total cost which includes disassembly, manufacturing, stock-out penalty, and inventory holding costs.

The demand for new items must be fully satisfied. However, stock-out is allowed for the refurbished items. When this happens, such demand can be either satisfied through downward substitution from new items, or is considered as a lost sale, which translates into a penalty cost in the objective function. Demand is time dependent over a discrete time horizon of T time periods. For each time period of the planning horizon ($t = 1...T$), the planner must decide on:

- the quantity to be disassembled and used for manufacturing of new products,
- the quantity of new parts to be purchased,
- the quantity of new products to be manufactured,
- the quantity of used products that will be refurbished or disassembled, and
- the quantity of new products that will be used for substitution

The quantity of products that can be collected and stored to enter the process is considered as decision variable in Section 3.1 and as input parameter in Sections 3.2.2 and 3.3.

Assumptions

The following assumptions are made to develop the model:

- All the accepted returns can be used for refurbishing or manufacturing.
- Both market-driven and waste stream systems (Guide, 2001) are used to represent how used items are obtained.
  - The market-driven approach relies on end-users to return their products to a firm specializing in the reuse of those products. End-users are motivated to return end-of-life products by financial incentives, such as deposit systems, credit toward a new unit, or cash paid for a specified level of quality. The level of returns is a decision variable in this case as suggested in the model of Section 3.1.
  - The waste stream approach relies on diverting discarded products from landfilling by making producers responsible for the collection and reuse of their products. In this situation, firms passively accept all product returns from the waste stream and the level of returns is a parameter as suggested in Sections 3.2 and 3.3.
- There are two alternatives (process flexibility) for refurbishing with a different costs. The first is to do the refurbishing in the specialized workshop with high setup but a reduced unit cost. The second is to integrate refurbishing with manufacturing with reduced setup cost but high unit cost and major (minor) setup times.

In the next sections, we will detail the following models:

1. A single grade (single quality level) model with shared or separate resources/capacities between refurbishing and manufacturing of new products and shared or separate inventory of returns for refurbishing and disassembly (Section 3.1).
2. A single grade model with uncertainty on returns and used product demands.
3. A multi grade (multi quality levels) model.

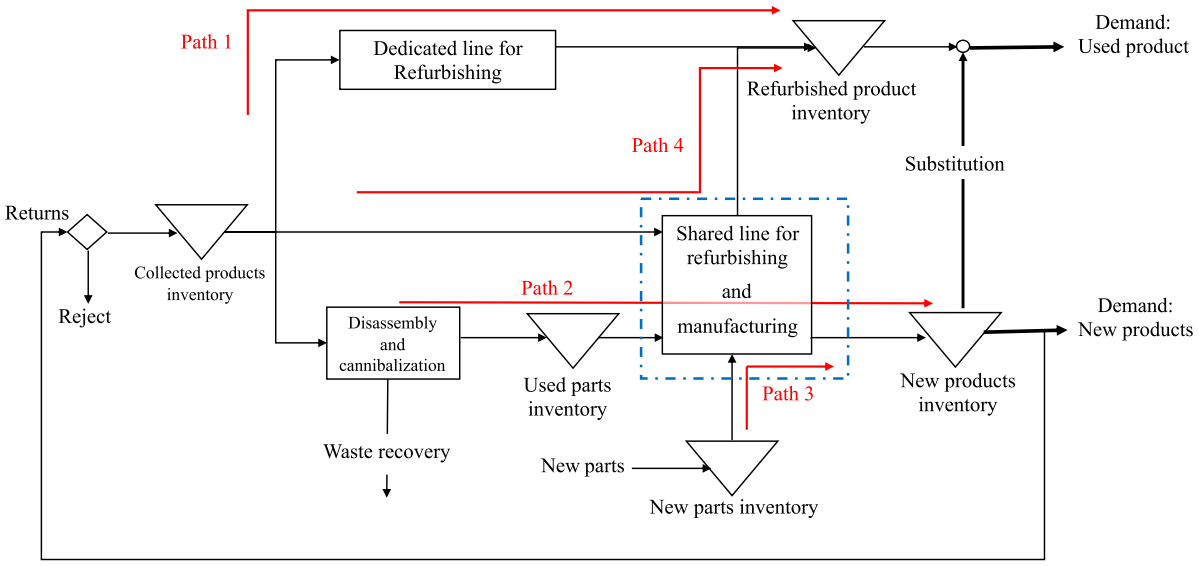


Fig. 1. The reference problem.

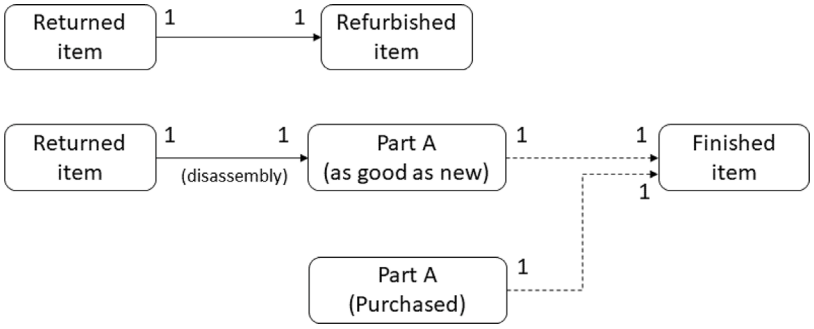


Fig. 2. The one-to-one relationships in the process. Dashed lines indicate the alternative for finished product to use Part A from external supplier or from a returned and disassembled product.

3.1. Single grade deterministic problem formulation

This section presents a MILP formulation for the first model, which is a deterministic single item problem. The process is as illustrated in Fig. 1. The amount of collected items is a decision variable as it is the case in market-driven systems for obtaining used products (Guide, 2001). Over a planning horizon of T discrete time periods ($t = 1..T$), the planner should satisfy all demand for new (manufactured) products d_t^a and eventually that of refurbished products d_t^r for each period t. If the latter is not satisfied, a stock-out penalty cost is incurred for each unsatisfied unit of the demand d_t^r. There is only one level of product quality/grade. Only downward substitution is allowed, that is, the demand for refurbished items can be partially or completely satisfied using new products at the same price of the refurbished items. The main objective of starting with such simplified model (compared to model in Section 3.3) is to analyze the complexity of the problem, assess the performance of the heuristic solution approach, and to analyze the impact of different parameters of the problem on some economic and environmental indicators. The notation used in the model is presented in Table 2.

Note that the unit refurbishing costs c^r and c^{ri} incurred on Path1 and Path4, respectively, correspond to all implicit costs associated with the refurbishing process (including, for example, the cleaning, upgrading, etc.). Similarly, the cost c^a includes all costs corresponding to the manufacturing (final assembly) process of the new product. Hence in Path2 of Fig. 1, the total unit cost $c^d + c^a$ of a new product, corresponds to all implicit costs associated with disassembling the part from a used product and manufacturing of final new product. Finally, on Path3, the total cost $c^a + c^n$ of a new product includes the unit cost of the purchased part together with all implicit costs associated with manufacturing of the new product.

This problem is formulated as a mixed integer linear programming formulation (MILP) as follows. This is referred to as Single Grade Deterministic Model (SGDM). This represents the problem shown in Fig. 1.

$$\begin{aligned}\textbf{Model SGDP}: \text{Maximize } Profit = &\sum_{t=1}^{T}\left(p^a d_t^a + p^r(d_t^r - L_t^r)\right) \\ &- \Big(\sum_{t=1}^{T}(h^c I_t^c + h^r I_t^r + h^d I_t^d + h^n I_t^n + h^a I_t^a) \\ &+ \sum_{t=1}^{T}(C^{sc} y_t^c + C^{sr} y_t^r + C^{sd} y_t^d + C^{sn} y_t^n + C^{sa} y_t^S) \\ &+ \sum_{t=1}^{T}(c^r q_t^r + c^{ir} q_t^{ir} + c^d q_t^d + c^n q_t^n + c^a q_t^a + c^c q_t^c) \Big) \end{aligned} \tag{1}$$

$$\text{Subject to: } I_t^c = I_{t-1}^c + q_t^c - q_t^r - q_t^{ir} - q_t^d \qquad t = 1..T \tag{2}$$

Table 2
Table of notations.

Indices	
t	Index for periods in the planning horizon, $t = 1, 2, \ldots, T$
Parameters	
d_t^a	Demand for new manufactured products.
d_t^r	Demand for refurbished (used) products.
h^c	Inventory holding cost for collected products.
h^r	Inventory holding cost for refurbished products.
h^d	Inventory holding cost for disassembled products.
h^n	Inventory holding cost for newly purchased parts.
h^a	Inventory holding cost for new manufactured products.
c^c	Collection cost for returned products.
c^r	Processing cost for refurbished products.
c^{ri}	Processing cost for refurbished products in integrated case.
c^d	Disassembly cost for disassembled products.
c^n	Purchasing cost for newly purchased parts.
c^a	Manufacturing cost for new manufactured products.
C^{sc}	Fixed setup cost for collection.
C^{sr}	Fixed setup cost for refurbishing.
C^{sd}	Fixed setup cost for disassembling.
C^{sn}	Fixed ordering cost for new parts.
C^{sa}	Fixed setup cost for manufacturing.
Q^r	Capacity of refurbishing expressed in time.
Q^a	Capacity of manufacturing expressed in time
Q^c	Inventory capacity for returned products.
T^M	Changeover time if manufacturing and refurbishing are processed in a shared line.
t^a	Unit time for manufacturing.
t^r	Unit time for refurbishing.
t^{ir}	Unit time for refurbishing in the shared line.
p^a	Price of new product.
p^r	Price of refurbished product.
M	A big number.
Variables	
I_t^{cr}	Inventory level of collected items for refurbishing in period t.
I_t^{cd}	Inventory level of collected items for disassembly in period t.
I_t^c	Inventory level of collected items in period t.
I_t^r	Inventory level of refurbished products in period t.
I_t^d	Inventory level of disassembled products in period t.
I_t^n	Inventory level of newly purchased parts in period t.
I_t^a	Inventory level of new manufactured products in period t.
q_t^{cr}	Quantity to be collected for refurbishing in period t.
q_t^{cd}	Quantity to be collected for disassembly in period t.
q_t^c	Quantity to be collected in period t.
q_t^r	Quantity to be refurbished in period t.
q_t^{ir}	Quantity to be refurbished in the shared line in period t.
q_t^{rd}	Quantity of refurbished items in period t used to satisfy demand for refurbished products.
q_t^u	Disassembled quantity used in the production in period t.
p_t^n	Quantity of new parts purchased in period t.
q_t^n	Quantity of new parts used in the production in period t.
q_t^a	Quantity of new items manufactured in period t.
q_t^s	Substitution quantity in period t.
L_t^r	Quantity to be loosed from d_t^r in period t.
y_t^{cr}	Binary variable equal to one if there is a collection in period t (items collected for refurbishing).
y_t^r	Binary variable equal to one if there is a refurbishment in period t.
y_t^d	Binary variable equal to one if there is a disassembly in period t.
y_t^n	Binary variable equal to one if there is a replenishment of new parts in period t.
y_t^a	Binary variable equal to one if there is a manufacturing in period t.
y_t^S	Binary variable equal to 1 if there is refurbishing or manufacturing in a shared line.
y_t^{ir}	Binary variable equal to 1 if the refurbishing is processed in a shared line.
y_t^M	Binary variable equal to 1 if the refurbishing and manufacturing are processed in a shared line.

$$I_t^r = I_{t-1}^r + q_t^r + q_t^{ir} - q_t^{rd} \qquad t = 1..T \tag{3}$$

$$I_t^d = I_{t-1}^d + q_t^d - q_t^u \qquad t = 1..T \tag{4}$$

$$I_t^n = I_{t-1}^n + p_t^n - q_t^n \qquad t = 1..T \tag{5}$$

$$I_t^a = I_{t-1}^a + q_t^a - q_t^s - d_t^a \qquad t = 1..T \tag{6}$$

$$I_t^c \le Q^c \qquad t = 1..T \tag{7}$$

$$q_t^{rd} + q_t^s = d_t^r - L_t^r \qquad t = 1..T \tag{8}$$

$$q_t^a = q_t^u + q_t^n \qquad t = 1..T \tag{9}$$

$$t^r q_t^r \le Q^r \times y_t^r \qquad t = 1..T \tag{10}$$

$$t^{ir} \times q_t^{ir} \le Q^a \times y_t^{ir} \qquad t = 1..T \tag{11}$$

$$t^a q_t^a \le Q^a \times y_t^a \qquad t = 1..T \tag{12}$$

$$t^a \times q_t^a + t^{ir} \times q_t^{ir} + T^M \times y_t^M \le Q^a \qquad t = 1..T \tag{13}$$

$$y_t^M \ge y_t^a + y_t^{ir} - 1 \qquad t = 1..T \tag{14}$$

$$y_t^S \ge y_t^{ir} \qquad t = 1..T \tag{15}$$

$$y_t^S \ge y_t^a \qquad t = 1..T \tag{16}$$

$$q_t^d \le M \times y_t^d \qquad t = 1..T \tag{17}$$

$$p_t^n \le M \times y_t^n \qquad t = 1..T \tag{18}$$

$$q_t^c \le M \times y_t^c \qquad t = 1..T \tag{19}$$

$$q_t^c, q_t^r, q_t^{ir}, q_t^{rd}, q_t^d, q_t^u, q_t^a, q_t^n, q_t^s, L_t^r, I_t^c, I_t^r, I_t^d, I_t^n, I_t^a \ge 0 \quad t = 1..T \tag{20}$$

$$y_t^c, y_t^r, y_t^d, y_t^a, y_t^n, y_t^{ir}, y_t^M, y_t^S \in \{0, 1\} \qquad t = 1..T \tag{21}$$

The objective (1) maximizes the total profit expressed as the difference between the sales revenue and the total costs composed of the variable and fixed purchasing/processing costs and inventory holding costs. Constraints (2)–(6) represent the inventory balance equations for collected items, refurbishing, disassembly, new parts, and manufacturing of new items, respectively. Constraints (7) represent inventory capacities of the returned products. Constraints (8) show that the demand for refurbished items can be partially satisfied using the output of refurbishing, substitution or lost. Constraints (9) indicate that the manufacturing of new items can use new purchased or disassembled parts. Constraints (10)–(12) link the continuous variables representing the quantities refurbished in the dedicated line, refurbished in the shared line, and manufactured to the corresponding binary variables, while respecting the capacities. If manufacturing and refurbishing are processed tougher in the shared line, constraints (13) and (14) indicate that a changeover time is considered. Constraints (15) and (16) link the binary variable y^S to y^a and y^{ir}, that means the setup cost is incurred each time the shared line is activated. Constraints (17)–(19) link the continuous variables representing the quantities purchased, collected and disassembled to the corresponding binary variables. *Big-M* represents an upper bound on the variables on the right-hand side of the constraints. Finally, constraints (20) and (21) are non-negativity and integrality constraints, respectively.

The main objective of starting with such simplified model (compared to model in Section 3.3) is to analyze the complexity of the problem, assess the performance of the heuristic solution approach, and to analyze the impact of different parameters of the problem on some economic and environmental indicators.

This model is used for the following purposes:

1. Assess the computational performance of the proposed heuristic approach.
2. Analyze the impact of different parameters of the problem on some economic and environmental indicators.

3.2. Single grade stochastic problem

The following model attempts to represent the situation with uncertainty on some demands (here d_t^r) and the amount of returns. This model corresponds to the waste stream approach for obtaining used items. In this situation, firms passively accept all product returns from the waste stream and the level of returns is a parameter. We still assume a single type of products as in Section 3.1. However, the returns amount is an input parameter instead of being a decision variable. There exist different approaches to tackle optimization problems with uncertainty on parameters. These include, for example, robust optimization and stochastic programming. The latter has received a lot of importance in solving many different optimization problems with uncertainty. Compared to robust optimization, stochastic programming assumes that the uncertain parameters follow a known probability distribution. For an introduction to stochastic programming and its applications, we refer to the book of Birge and Louveaux (2011).

In stochastic programming, the two-stage formulation has been widely used in the literature. Its basic idea is that decisions should be based on data available at the time the decisions are made. Such decisions should not depend on future observations. Hence, decisions are made in two stages and the observation of some random event takes place in between. The first set of decisions, that must be taken before the observation of any of the uncertain elements in the problem, are called first-stage decisions. The second set of decisions, that can be taken after the occurrence of uncertain events, are called second stage decisions.

Return and demand uncertainties are the common uncertain parameters in the literature of production planning with returns (Ferguson et al., 2009; Azizi et al., 2020). As problems with uncertain parameters represented with continuous distribution are computationally challenging, scenarios are typically used to approximate the underlying continuous distributions. Each scenario is a set of discrete values showing returns from the primary markets and demands in the secondary markets (Azizi et al., 2020).

In this section, a two-stage stochastic programming is proposed where demand for refurbished products d^r and the amount of returned products q^c are random parameters. In this case, the first-stage variables are used to define the quantities to be manufactured, refurbished, disassembled, and purchased (q^a, q^r, q^d, p^n) and their respective setups (y^a, y^r, y^d, y^n). The second stage refers to the inventory levels (I^a, I^r, I^d, I^n, I^c), the quantities used for satisfying demand for refurbished products (q^s, q^{rd}) and the stock-out level L^r.

However, the quality of the uncertain information may not be good enough to be expressed as probability distributions; even if the probability distributions are available, it could be difficult to reflect them in large scale models. Consequently, interval parameters are introduced into the stochastic programming framework to facilitate communication of the uncertainties into the optimization process, that can be reasonably well described by a finite set of realizations called scenarios. For each scenario, there is a corresponding probability of occurrence.

3.2.1. Two-stage stochastic model

We assume that the two parameters, amount of returned items and the demand for refurbished items, are random variables described by a finite set of Ξ scenarios. The scenario subscript $\xi = 1...\Xi$ is added to all decision variables of second stage to represent a scenario with corresponding probability π_ξ.

$$\begin{aligned}\textbf{Model TSSM}: \text{Maximize } Profit = &\sum_{t=1}^{T} p^a d_t^a \\ &- \sum_{t=1}^{T}(C^{sr} y_t^r + C^{sd} y_t^d + C^{sn} y_t^n + C^{sa} y_t^S) \\ &- \sum_{t=1}^{T}(c^r q_t^r + c^{ir} q_t^{ir} + c^d q_t^d + c^n q_t^n + c^a q_t^a) \\ &- \sum_{t=1}^{T}\sum_{\xi=1}^{\Xi} \pi_\xi (h^c I_{t\xi}^c + h^r I_{t\xi}^r + h^d I_{t\xi}^d + h^n I_{t\xi}^n + h^a I_{t\xi}^a + c^c q_{t\xi}^c - p^r (d_{t\xi}^r - L_{t\xi}^r))\end{aligned} \tag{22}$$

Subject to: (9)–(19).

$$I_{t\xi}^c = I_{(t-1)\xi}^c + q_{t\xi}^c - q_t^r - q_t^{ir} - q_t^d \qquad t = 1..T, \quad \xi = 1..\Xi \tag{23}$$

$$I_{t\xi}^r = I_{(t-1)\xi}^r + q_t^r + q_t^{ir} - q_{t\xi}^{rd} \qquad t = 1..T, \quad \xi = 1..\Xi \tag{24}$$

$$I_{t\xi}^d = I_{(t-1)\xi}^d + q_t^d - q_t^u \qquad t = 1..T, \quad \xi = 1..\Xi \tag{25}$$

$$I_{t\xi}^n = I_{(t-1)\xi}^n + p_t^n - q_t^n \qquad t = 1..T, \quad \xi = 1..\Xi \tag{26}$$

$$I_{t\xi}^a = I_{(t-1)\xi}^a + q_t^a - q_t^s - d_t^a \qquad t = 1..T, \quad \xi = 1..\Xi \tag{27}$$

$$I_{t\xi}^c \le Q^c \qquad t = 1..T, \quad \xi = 1..\Xi \tag{28}$$

$$q_{t\xi}^{rd} + q_{t\xi}^s = d_{t\xi}^r - L_{t\xi}^r \qquad t = 1..T, \quad \xi = 1..\Xi \tag{29}$$

$$q_t^r, q_t^{ir}, q_t^d, q_t^u, q_t^a, q_t^n \ge 0 \qquad t = 1..T \tag{30}$$

$$q_{t\xi}^{rd}, q_{t\xi}^s, L_{t\xi}^r, I_{t\xi}^c, I_{t\xi}^r, I_{t\xi}^d, I_{t\xi}^n, I_{t\xi}^a \ge 0 \qquad t = 1..T, \quad \xi = 1..\Xi \tag{31}$$

$$y_t^r, y_t^d, y_t^a, y_t^n, y_t^{ir}, y_t^M, y_t^S \in \{0,1\} \qquad t = 1..T \tag{32}$$

The objective (22) aims to maximize the expected total profit over the planning horizon. It consists of two parts: the first is linked to the first stage variables, while the second one is related to the second stage variables. Constraints (23)–(29) are equivalent to constraints (2)–(8) involving the second stage variables. Constraints (30) and (32) are non-negativity and integrality constraints for the first stage variables, and (31) are non-negativity constraints for the second stage variables.

3.2.2. Risk-averse two-stage stochastic model

The main idea of a risk averse stochastic models is to reduce the variability of specific outcome realizations over the scenarios by introducing a variable that measures the robustness of the model. Common risk measures used include risk deviation that evaluate the dispersion of the second part related to the second stage variables in the objective function.

Among the different risk deviation measures, we adopt the well-known upper partial mean that is defined as the expected value of the positive deviation (Mulvey et al., 1995), and we introduce a new decision variable δ_ξ to evaluate the deviation between the total profit of the scenario ξ and the expected total profit. This variable is inserted into constraints and penalized into the objective function by a parameter $\lambda > 0$. A similar approach was proposed in Macedo et al. (2016) on a lot sizing problem with remanufacturing.

The corresponding TSSM risk-averse model is presented as follows:

$$\begin{aligned}\textbf{Model RATSSP}: \text{Maximize } Profit = &\sum_{t=1}^{T} p^a d_t^a \\ &- \sum_{t=1}^{T}(C^{sr} y_t^r + C^{sd} y_t^d + C^{sn} y_t^n + C^{sa} y_t^S) \\ &- \sum_{t=1}^{T}(c^r q_t^r + c^{ir} q_t^{ir} + c^d q_t^d + c^n q_t^n + c^a q_t^a) \\ &- \sum_{t=1}^{T}\sum_{\xi=1}^{\Xi} \pi_\xi (h^c I_{t\xi}^c + h^r I_{t\xi}^r + h^d I_{t\xi}^d + h^n I_{t\xi}^n + h^a I_{t\xi}^a + c^c q_{t\xi}^c - p^r (d_{t\xi}^r - L_{t\xi}^r)) \\ &- \lambda \sum_{\xi=1}^{\Xi} \pi_\xi \delta_\xi \end{aligned} \tag{33}$$

Subject to: (9)–(19) and (23)–(32).

$$\begin{aligned}\delta_\xi \geq &\sum_{t=1}^{T}(h^c I_{t\xi}^c + h^r I_{t\xi}^r + h^d I_{t\xi}^d + h^n I_{t\xi}^n + h^a I_{t\xi}^a + c^c q_{t\xi}^c - p^r (d_{t\xi}^r - L_{t\xi}^r)) \\ &- \sum_{t=1}^{T}\sum_{\xi'=1}^{\Xi} \pi_{\xi'} (h^c I_{t\xi'}^c + h^r I_{t\xi'}^r + h^d I_{t\xi'}^d + h^n I_{t\xi'}^n + h^a I_{t\xi'}^a \\ &+ c^c q_{t\xi'}^c - p^r (d_{t\xi'}^r - L_{t\xi'}^r)) \qquad \xi = 1..\Xi \end{aligned} \tag{34}$$

$$\delta_\xi \geq 0 \qquad \xi = 1..\Xi \tag{35}$$

The objective (33) maximizes the profit while minimizing the risk represented by the expected deviation. By increasing λ, we obtain solutions with low expected profit in favor of lower variability. Constraints (34) and (35) state the definition of the upper partial mean deviation.

Profit maximization and risk minimization are two conflicting objectives. They are jointly optimized in the objective function through the parameter λ. The latter can be considered as the *risk aversion control parameter*. From a managerial point of view, higher values of λ correspond to highly risk averse decision makers who prefer to minimize risk at the expense of reducing the total profit.

3.3. Multi grade deterministic problem formulation

Vertical differentiation is a strategy that has been adopted by companies for a long time. Vertical differentiation is related to offering products with different quality levels but offering essentially the same functionalities. The products (models) can be classified in this case as high, medium, and low-quality; or equivalently as premium, standard and basic models (Randall et al., 1998). An example, is the Samsung Galaxy Note 10 Plus, Note 10, and Note 10 Lite. The three models offer basically the same functionalities but differ in some performance and finishing levels. Another example is the Nikon SLR 35 mm cameras N50, N70, and N90 models, which again differ only in quality level (Randall et al., 1998). Such differentiation is made to satisfy customers' different needs at different price levels.

The quality level or grade should be taken considered when returned items are received. A priori, a product of a given quality level should be refurbished at the same level of quality. For example, a premium returned product should be refurbished to satisfy the demand for a premium used product. It is important to highlight the difference between the quality grading concept related to vertical differentiation as defined above from the grading of returns as an operational decision where returned items are tested for any technical issues are categorized (graded) based on the required efforts and costs to refurbish or remanufacture them (Ferguson et al., 2009). Though such grading systems have proven their importance to increase profitability as indicated in the paper of Ferguson et al. (2009), this option is not considered in the current work as our focus is on the analysis of upgrading and downgrading of different product models (grades).

Upgrading and downgrading occur mainly in refurbishing. Upgrading a low-quality product (a basic model, for example) consists of replacing worn-out parts of such model with parts extracted from a returned product of high quality (a premium model, for example). Downgrading occurs in the opposite direction where parts from a returned low-quality product are used in a high quality product. These operations are possible because the vertical differentiation usually allows some level of compatibility between spare parts of the products. For example, an S-pen can be used in the Samsung Galaxy Note 10 Lite, 10, and 10 Plus models, though the S-pen of the Galaxy Note 10 Plus might have underdone higher quality inspection efforts compared to the one bundled with the Galaxy Note 10 Lite.

In this section, we consider a Multi Grade Deterministic Model (MGDM) to consider a new product with n quality grades. Each grade has its own associated returns, refurbished products, and new parts, leading to n distinct cost structures. Each returned unit of quality $i = 1, \ldots, n$ has several recovery options, namely:

- refurbished to product of quality i,
- refurbished to product of lower quality than i (downgrading),
- refurbished to product of higher quality than i (upgrading), with an additional cost c_{ij}^{up} for upgrading, or
- disassembled to extract a single key part to be used in manufacturing of a new product of grade quality i.

In the MGDM model, we employ the same notations used in the previous models with the addition of subscript i to denote a grade level. The variable q_{ijt}^r (respectively, q_{ijt}^{ir}) represents the quantity of returns of grade quality i refurbished on the dedicated line (respectively, on the shared line) to obtain the refurbished product of quality j in time period t.

$$\begin{aligned}\textbf{Model MGDP}: \text{Maximize } Z = &\sum_{t=1}^{T}\sum_{i=1}^{n}\left(p_i^a d_{it}^a + p_i^r\left(d_{it}^r - L_{it}^r\right)\right) \\ &- \left(\sum_{i=1}^{n}\sum_{t=1}^{T}\left(h_i^c I_{it}^c + h_i^r I_{it}^r + h_i^d I_{it}^d + h_i^n I_{it}^n + h_i^a I_{it}^a\right)\right. \\ &\left. + \sum_{t=1}^{T}\left(C^{sr} y_t^r + C^{sd} y_t^d + C^{sn} y_t^n + C^{sa} y_t^S\right)\right. \\ &\left. + \sum_{i=1}^{n}\sum_{t=1}^{T}\left(c_i^r \sum_{j}^{n} q_{jit}^r + c_i^{ir} \sum_{j}^{n} q_{jit}^{ir} + c_i^d q_{it}^d + c_i^n q_{it}^n + c_i^a q_{it}^a + c_i^c q_{it}^c \right.\right. \\ &\left.\left. + \sum_{j>i}^{n} c_{ij}^{up}(q_{ijt}^r + q_{ijt}^{ir})\right)\right) \end{aligned} \tag{36}$$

Subject to: $I_{it}^c = I_{it-1}^c + q_{it}^c - q_{it}^d - \sum_{j=1}^{n}(q_{ijt}^r + q_{ijt}^{ir}) \qquad i = 1..n\,, \quad t = 1..T$ (37)

$$I_{it}^r = I_{it-1}^r + \sum_{j=1}^{n}(q_{jit}^r + q_{jit}^{ir}) - q_{it}^{rd} \qquad i = 1..n\,, \quad t = 1..T \tag{38}$$

$$I_{it}^d = I_{it-1}^d + q_{it}^d - q_{it}^u \qquad i = 1..n\,, \quad t = 1..T \tag{39}$$

$$I_{it}^n = I_{it-1}^n + p_{it}^n - q_{it}^n \qquad i = 1..n\,, \quad t = 1..T \tag{40}$$

$$I_{it}^a = I_{it-1}^a + q_{it}^a - q_{it}^s - d_{it}^a \qquad i = 1..n\,, \quad t = 1..T \tag{41}$$

$$\sum_{i=1}^{n} I_{it}^c \leq Q^c \qquad t = 1..T \tag{42}$$

$$q_{it}^{rd} + q_{it}^s = d_{it}^r - L_{it}^r \qquad i = 1..n\,, \quad t = 1..T \tag{43}$$

$$q^a_{it} = q^u_{it} + q^n_{it} \qquad i = 1..n\,, \quad t = 1..T \tag{44}$$

$$\sum_{i=1}^{n}\sum_{j=1}^{n} t^r_i q^r_{jit} \le Q^r y^r_t \qquad t = 1..T \tag{45}$$

$$\sum_{i=1}^{n}\sum_{j=1}^{n} t^{ir}_i q^{ir}_{jit} \le Q^a y^{ir}_t \qquad t = 1..T \tag{46}$$

$$\sum_{i=1}^{n} t^a_i q^a_{it} \le Q^a y^a_t \qquad t = 1..T \tag{47}$$

$$\sum_{i=1}^{n} t^d_i q^d_{it} \le M y^d_t \qquad t = 1..T \tag{48}$$

$$\sum_{i=1}^{n} p^n_{it} \le M y^n_t \qquad t = 1..T \tag{49}$$

$$\sum_{i=1}^{n} t^a_i q^a_{it} + \sum_{i=1}\sum_{j=1} t^{ir}_i q^{ir}_{jit} + T^M \times y^M_t \le Q^a \qquad t = 1..T \tag{50}$$

$$y^M_t \ge y^a_t + y^{ir}_t - 1 \qquad t = 1..T \tag{51}$$

$$y^S_t \ge y^{ir}_t \qquad t = 1..T \tag{52}$$

$$y^S_t \ge y^a_t \qquad t = 1..T \tag{53}$$

$$q^r_{jit}, q^{ir}_{ijt}, q^{rd}_{it}, q^d_{it}, q^u_{it}, q^a_{it}, q^n_{it}, q^s_{it}, I^c_{it}, I^r_{it}, I^d_{it}, I^n_{it}, I^a_{it}, L^r_{it} \ge 0 \;\; i,j = 1..n\,, \quad t = 1..T \tag{54}$$

$$y^r_t, y^d_t, y^a_t, y^n_t, y^{ir}_t \in \{0,1\} \qquad t = 1..T \tag{55}$$

The objective (36) aims to maximize the total profit over the planning horizon and grade qualities. It is expressed as the difference between the sales revenue and the total costs. Note that the upgrading cost is incurred when a returned item of grade i is upgraded to an item of grade j with an additional cost c^{up}_{ij}. Constraints (37)–(41) represent the inventory balance equations. Constraints (42) represent inventory capacities of the returned products. Constraints (43) show that the demand for refurbished items can be fully or partially satisfied and constraints (44) indicate that the new purchased or disassembled parts of grade i can be used in manufacturing of new items of the same grade. Constraints (45)–(49) link the continuous variables to the corresponding binary variables, while respecting the capacities. Constraints (50) impose a processing capacity limit on the shared line. Constraints (51)–(53) link the binary variables.

4. A heuristic solution approach

It should be noted that this problem is NP-hard. This is a generalization of the One Warehouse Multi Retailer (OWMR) problem, which was proven to be NP-hard by Arkin et al. (1989). For this reason, heuristic techniques are adopted to solve it. With the growing performance of MILP-solvers, MILP-based heuristics have been employed to solve MILP models. Among the existing MILP-based heuristics, we have chosen relax-and-fix heuristic and fix-and-optimize heuristic.

The main idea of relax-and-fix heuristics (RF) consists of decomposing a MILP model to iteratively solve sub-problems with a reduced number of variables with integrality constraints. In the case of a MILP with binary variables, the latter are partitioned into three subsets in each iteration of the procedure. The first subset is set to fixed binary values (from previous iterations). The second subset is relaxed to take arbitrary continuous values between 0 and 1. The third subset is restricted with integrality constraints to take only values 0 or 1. This reduces the number of binary variables in the sub-problem of the current iteration. The solution time for a sub-problem is very small. Relax-and-fix heuristics were developed for different production planning and lot sizing problems. More details of the relax-and-fix heuristic for lot sizing problems can be found in Pochet and Wolsey (1991).

Fix-and-optimize heuristic (FO) optimizes a subset of variables by solving a series of sub-problems in an iterative manner. A sub-problem is the reduced form of the original problem where a set of decision variables are fixed to integer values (sometimes randomly) whereas the remaining set are left to be optimized. Like the relax-and-fix heuristics, the FO heuristics are known for their ease of implementation, and thus, have been commonly adopted in various lot sizing problems over the last decade. For more information on the fix-and-optimize heuristic, the reader can refer to the studies of Pochet and Wolsey (1991) and Sahling et al. (2009).

The next sections present the two phases of a *Time-based relax-and-fix and fix-and-optimize heuristic* (TRF-FO). In the first phase, a good feasible solution using time-based relax-and-fix construction heuristic (TRF) presented in Section 4.1. In the second phase, an improvement process is executed on this solution using fix-and-optimize heuristic (FO) as shown in Section 4.2.

4.1. Time-based relax-and-fix heuristic (TRF)

The time-based relax-and-fix heuristic is based on the decomposition of the planning horizon into three intervals or windows of periods, which represent internally rolled horizons. In the first window of periods, which we call W^F, the binary variables are fixed to their optimal value from the solution of a MILP sub-problem solved in the previous iterations. In the second window of periods, called W^I, the setup variables are forced to be binary. The binary variables in the last window, W^R, are relaxed and take any real value between 0 and 1. The length of each window is determined using to two parameters, α and β, where α and β are integer values and $\beta < \alpha$. Then, iteratively, the windows will be rolling from the beginning of the planning horizon to the end as illustrated in Fig. 3. In the first iteration, a MILP sub-problem forces integrality constraints on the binary variables in the first α periods (where $\alpha < T$), and all other binary variables are relaxed. In the second iteration, binary variables in periods $t = 1$ to β are fixed to their optimal values from the solution of the MILP sub-problem solved in the first iteration. The setup variables from period $\beta + 1$ to period $\beta + \alpha$ are then forced to be binary and relaxed from period $\beta + \alpha + 1$ to period T.

In each iteration k, the three windows W^F, W^I, and W^R are defined as follows:

$$W^F = [1, k \times \beta]$$
$$W^I = [k \times \beta + 1, k \times \beta + \alpha]$$
$$W^R = [k \times \beta + \alpha + 1, T]$$

For $k = 0$, W^F is empty, $W^I = [1, \alpha]$, and $W^R = [\alpha + 1, T]$.

Where α represents the length of window W^I, and β is the forward sliding step of window W^I.

From iteration k to iteration $k+1$, the following β periods are added to the window W^F. Window W^I is shifted forward by β periods and the first β periods are eliminated from window W^R.

Let $\overline{y}^{(k)}_t$ be the integer value of y_t found at iteration k.

The following relax-and-fix lot sizing ($RFLS$) model is solved at each iteration k:

Model: $TRFLS$

Maximize $Profit$ (See Eq. (1))

subject to (2)–(20) and:

$$y^b_t = \overline{y}^{b(k-1)}_t \qquad t \in W^F, \;\; b = \{c, r, d, n, a\} \tag{56}$$

$$y^b_t \in \{0,1\} \qquad t \in W^I, \quad b = \{c, r, d, n, a\} \tag{57}$$

$$y^b_t \in [0,1] \qquad t \in W^R, \quad b = \{c, r, d, n, a\} \tag{58}$$

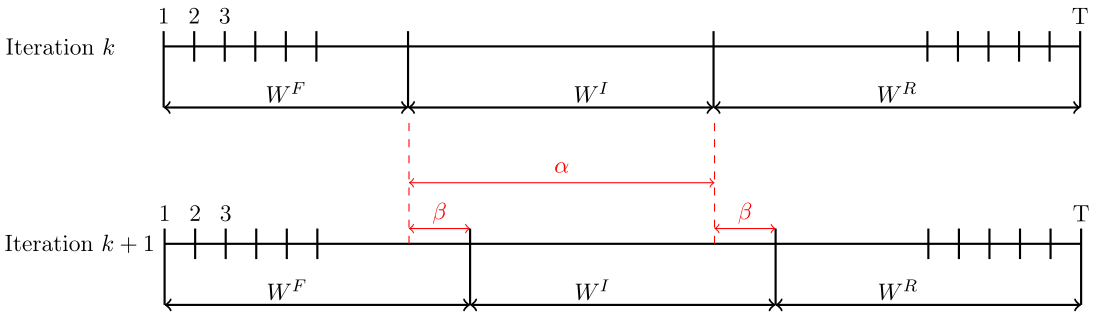


Fig. 3. Forward sliding of windows in TRF algorithm.

The sub-problems in each iteration of the TRFLS heuristic are solved using an MIP solver (CPLEX in this case), and the solution determines the production levels in each period in W^I.

The pseudo code of the TRF heuristic is presented in Algorithm 1.

Algorithm 1 Pseudocode of time-based relax-and-fix heuristic (TRF)

$k = 0$
$W^F = \emptyset$
$W^I = [1, \alpha]$
$W^R = [\alpha + 1, T]$
$Solve\ RFLS$
$k = k + 1$
while $k \times \beta + \alpha < T$ **do**
  $W^F = [1, k \times \beta]$
  $W^I = [k \times \beta + 1, k \times \beta + \alpha]$
  $W^R = [k \times \beta + \alpha + 1, T]$
  $Solve\ RFLS$
  $k = k + 1$
end while
$W^F = [1, k \times \beta]$
$W^I = [k \times \beta + 1, T]$
$W^R = \emptyset$
$Solve\ RFLS$

4.2. Fix-and-optimize heuristic (FO)

The FO heuristic (summarized in Algorithm 2) starts from an initial solution and considers all integrality constraints at every iteration (i.e. no binary variables are relaxed). It is based on the decomposition of the binary variables into two subsets. The first subset is made of variables whose values are fixed to 0 or 1. The second subset is composed of free binary variables (with integrality constraints). There are different ways to choose the fixed and the to-be-optimized variables. The most well known approach is a time-based approach where the planning horizon is divided into two windows of periods, W^F and W^I, which are internally rolled horizons. W^I is the subset of periods where the setup variables are to be re-optimized. The length of W^I window is equal to parameter γ which can take its values arbitrarily between $[\gamma min, \gamma max]$. The W^F subset includes the remaining periods where the binary variables are fixed to their optimal value from the solution of an MILP sub-problem solved in the previous iterations. An illustration of the sliding of window W^I is shown on Fig. 4. Starting with an initial solution S^0 which can be obtained by a construction heuristic (TRF heuristic in case of TRF-FO approach), the first iteration solves a MILP sub-problem where binary variables in the first γ periods are reoptimized while the remaining variables are fixed. In the next iteration, if there is no improvement in the objective function compared to the best solution so far, then we slide the W^I window by γ periods, update W^F and solve a MILP sub-problem. If there is an improvement in the objective function, the best solution is updated as the current solution. Then we restart W^I window from the beginning of the horizon with a new value of γ (chosen randomly between $[\gamma^{min}, \gamma^{max}]$) and we reset iteration counter to zero. The FO heuristic stops when there is no improvement and W^I window reaches the end of the horizon. The MILP sub-problem solved in FO heuristic (we call it $FOLS$) is the same as $TRFLS$ without constraints (58).

Algorithm 2 Pseudocode of fix-and-optimize heuristic

$S^* = S^0$
$\gamma = U[\gamma^{min}, \gamma^{max}]$
$k = 0$
while $k \times \gamma < T$ **do**
  $W^I = [k \times \gamma + 1, (k + 1) \times \gamma]$
  $W^F = [1, T] \backslash W^I$
  $Solve\ FOLS$
  if $f(S) < f(S^*)$ **then**
    $S^* = S$
    $\gamma = U[\gamma^{min}, \gamma^{max}]$
    $k = 0$
  else
    $k = k + 1$
  end if
end while

5. Numerical experiments

The numerical experiments are divided into two sections: the first section focuses on computational performance of the exact (solver) approach and the heuristic procedure. The second section provides managerial insights.

The computational experiments analyze the complexity of the single grade deterministic problem (SGDM) described in Section 3 under different as a function of different problem parameters. The experiment have two main objectives:

- For small size instances, the numerical performance of the proposed heuristic is evaluated by comparing the best solution with the optimal solution obtained by a commercial MILP solver.
- For larger size instances, where the MILP solver was not able to find any optimal solutions in reasonable CPU time, a limited time is fixed to the solver and the results obtained are compared with those obtained by heuristic approach.

The managerial insight experiments study the environmental and economic impact of the key parameters. The objective of this experiments is threefold:

- For the single grade deterministic model (SGDM), the impacts of variations in unit costs on different indicators were analyzed. The indicators include profit, collection efficiency, substitution, and lost sales. This analysis allows decision makers interested in hybrid refurbishing and manufacturing systems to identify conditions and parameters that impact such performance indicators.

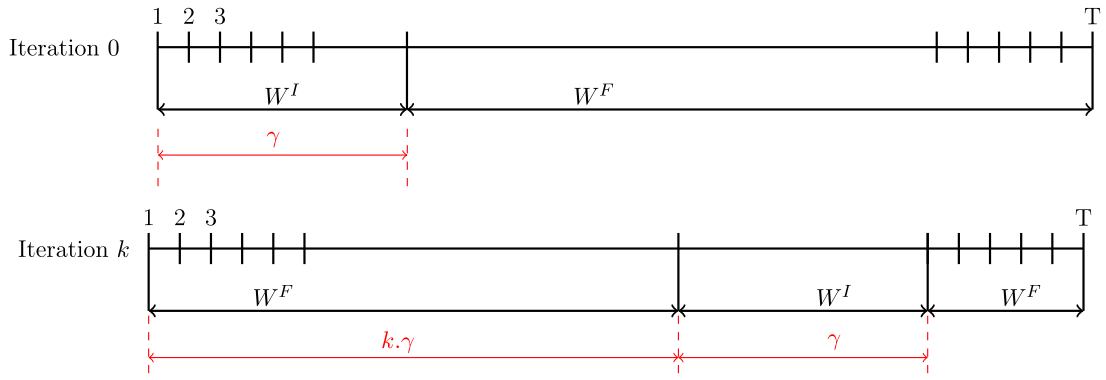


Fig. 4. Forward sliding of windows in FO algorithm.

- For the Single Grade Stochastic Model (SGSM), the impact of uncertainty and risk aversion are analyzed. This enables decision-makers to make more informed and robust decisions by explicitly considering uncertainty and risk in their planning process.
- For the multi grade deterministic model (MGDM), different flows are analyzed including the amounts upgraded and downgraded between and illustrated using *Sankey diagrams*.

The heuristics were implemented in GAMS and solved using the commercial solver CPLEX 12.4.

All experiments were run using a PC with AMD Ryzen 7 PRO 5850U, 1.90 GHz, 16 GB RAM, under a Windows 10 professional operating system.

5.1. Data generation

The key parameters such as the length of the planning horizon T, the demand of new products d^a, the setup costs for manufacturing C^{sa}, and the holding cost for new manufactured products h^a are generated from Teunter et al. (2006), among others. The other parameters are generated according to the key parameters, within a reasonable range, and to make sure that all processes (collection, disassembly, refurbishing, refurbishing in shared line, purchasing) of new parts, manufacturing and substitution take place.

The description of the default values (nominal value) of all parameters is discussed as follows:

- The length of the planning horizon is $T = 12$.
- Manufacturing and refurbishing capacities, Q^a and Q^r are fixed to 200 and 100, respectively.
- Demands d^a and d^r are a ratio of capacities and have been generated randomly from a normal distribution with a mean μ^a, μ^r and a standard deviations σ^a, σ^r respectively. $\mu^a = 0.5 \times Q^a$, $\mu^r = 0.5 \times Q^r$. $\sigma^a = 0.15 \times \mu^a$, $\sigma^r = 0.15 \times \mu^r$.
- The holding cost h^a is set equal to 1, and the manufacturing cost $c^a = h^a/0.15$.
- The others unit costs (collection, disassembling, refurbishing, and purchasing) are a percentage of manufacturing cost as follows: $c^c = 0.25 \times c^a$, $c^d = 0.1 \times c^a$, $c^r = 0.7 \times c^a$, $c^{ir} = 0.9 \times c^a$, $c^n = 0.5 \times c^a$. The choice of percentages is made within a reasonable range, aiming to create a conflict between disassembling and purchasing of new parts, and between refurbishing and substitution, i.e. the sum of the unit costs of collection and disassembling should be lower than the purchasing unit cost, and likewise, the same of unit costs of collection and refurbishing should be lower than the combined costs of purchasing and manufacturing.
- The other holding costs at each stage are assumed to be a percentage of the total unit costs until this stage, i.e. $h^c = 0.1 \times c^c$, $h^r = 0.1 \times (c^c + c^r)$, $h^d = 0.1 \times (c^c + c^d)$, $h^n = 0.1 \times c^n$.

Table 3
Variation values of parameters in computational experiments.

Parameter	D_{Val}	D_{Var}	Setup	Changeover
Low	$0.5 \times \mu$	$0.5 \times \sigma$	$0.5 \times C^s$	$0.5 \times T^M$
High	$1.5 \times \mu$	$1.5 \times \sigma$	$4 \times C^s$	$1.5 \times T^M$

- The setup cost for manufacturing C^{sa} is set to 500, and the others setup costs are generated according to C^{sa} as follows: $C^{sr} = 0.7 \times C^{sa}$, $C^{sd} = 0.8 \times C^{sa}$, $C^{sn} = 0.2 \times C^{sa}$, $C^{sc} = 0.8 \times C^{sa}$. The setup cost for purchasing should be lower than the setup cost of collection and disassembling, otherwise, there will never be a purchase.
- The price of new products is fixed to $P^a = 4.5 \times c^a$; i.e. $P^a = 30$, and does not influence the value of the optimal solution but it is useful to express the net profit (total sales minus total cost) . Since lost sales of refurbished products are allowed, then price of refurbished products P^r has an influence on the optimal solution, and its defaults value is set to $3 \times c^a$; i.e. $P^r = 0.66 \times P^a$, which is a realistic value.
- The changeover time T^M is estimated as $15\% \times Q^a$, and the unit time t^a, t^r and t^{ir} for manufacturing and refurbishing in the shared line, respectively, are all equal to 1.

For each combination of the parameters, five instances were randomly generated.

5.2. Performance of the heuristic

The performance of the heuristic described above (TRF-FO) was done on the Single Grade Deterministic Model under two levels (low or high) of the parameters: demands, setup costs, and changeover times. The level of values of each parameter is given as a ratio of its nominal value as shown in Table 3. Demand value (D_{Val}) is a variation in the mean of demand μ whereas demand variability (D_{Var}) is a variation in a standard deviation σ. All other parameters are at their nominal values.

In the case of small-size instances with $T = 12$ time periods, the performance is measured using the GAP between the value of the solutions obtained by the heuristic (z^H) and the optimal solution (z^{OPT}) obtained using the MILP solver CPLEX. The GAP was computed using the formula:

$$GAP = 100 \times \frac{z^{OPT} - z^H}{z^{OPT}}$$

For large size instances ($T = 24$ time periods), where the MILP solver CPLEX was not able to find any optimal solutions in reasonable CPU time, the performance of the heuristic was measured by calculating the distance of its solution from the best solution found by CPLEX within a time limit of 100 s

Table 4
Performance of the TRF-FO heuristic for small instances ($T = 12$).

Parameter		Case 1: Base case			Case 2: No line sharing			Case 3: Zero changeover time		
		CPLEX CPU (s)	TRF-FO CPU (s)	TRF-FO GAP (%)	CPLEX CPU (s)	TRF-FO CPU (s)	TRF-FO GAP (%)	CPLEX CPU (s)	TRF-FO CPU (s)	TRF-FO GAP (%)
D_{Var}	Low	6.90	1.09	0.35	6.92	1.33	0.24	4.08	1.27	0.13
	High	5.11	0.99	0.30	5.02	1.30	0.13	2.73	1.18	0.30
D_{Val}	Low	1.47	0.86	0.89	1.46	0.97	0.89	1.16	0.89	1.23
	High	2.29	0.90	0.16	2.28	1.18	0.02	1.21	1.07	0.12
SETUP	Low	1.11	0.88	0.21	1.11	1.08	0.11	0.98	1.06	0.11
	High	2.64	0.73	0.57	2.64	0.92	0.00	1.22	0.88	0.00
Changeover time	Low	6.83	1.08	0.41	6.77	1.32	0.26	–	–	–
	High	10.60	1.04	0.31	10.66	1.31	0.10	–	–	–
Average		4.62	0.95	0.40	4.61	1.18	0.22	1.90	1.06	0.32

It should be noted that the performance of the heuristic strongly depends on the choice of its parameters. For the construction relax-and-fix heuristic TRF, the solution quality improves as α increases and β decreases, but it will require longer computational times. To have a good compromise between quality and CPU time, several tests were performed over randomly chosen instances using different combinations of (α, β).

We found that a combination $(\alpha, \beta) = (6,4)$ gave the best compromise on most tests. The improvement heuristic FO is characterized by a γ parameter which can take its values arbitrarily between $[\gamma^{min}, \gamma^{max}]$. We recall that if there is an improvement in the solution, we restart the heuristic with a new value of γ. Notice that when γ increases, both the performance and the running time of the heuristic increase. To set the values of γ^{min} and γ^{max}, several tests were performed. We found a good compromise between the CPU time and solution quality where $\gamma^{min} = 4$ and $\gamma^{max} = 6$.

The numerical study is conducted to evaluate the performance of the TRF-FO heuristic on the following cases:

(*i*) *Case 1*: The Single Grade Deterministic Model (SGDM).

(*ii*) *Case 2*: The Single Grade Deterministic Model (SGDM) while preventing sharing of the new product line with refurbishing. In this case, the variable y^{ir} is set equal to 0.

(*iii*) *Case 3*: The Single Grade Deterministic Model (SGDM) with zero changeover time between refurbishing and new item production on the shared line. In this case, T^M is set equal to 0.

For instances with $T = 12$ periods, Table 4 shows the CPU times of CPLEX and the heuristic CPU times and gaps from optimum in the three cases introduced previously. These results indicate that the heuristic is rather efficient with an average CPU time of 1 s and a gap of 0.3% from optimum. The solver converges faster when the changeover time is set to zero (Case 3). Results in both Cases 1 and 2 indicate that the changeover time has an important impact on the performance of the solver. CPLEX takes up to 10 s on average when the changeover time is high. Overall, the quality of the solutions obtained by the heuristic is rather stable though we can observe the following.

Demand variability does not influence the performance of the heuristic in Case 1. In Case 2 lower variability results in higher gaps while it results in better solutions in Case 3. In all cases, larger average demands and changeover times result in better quality solutions of the heuristic contrary to the results obtained using CPLEX.

Table 5 summarizes the results obtained on problems with a planning horizon of 24 time periods where CPLEX maximum CPU time was set to 100 s. The average gap between heuristic solutions and those obtained using CPLEX is 0.2%. This is an excellent result knowing that the heuristic takes on average 4 s on these instances.

5.3. Managerial insights

The objective in these experiments is to study the impact of different problem parameters on a set of Key Performance Indicators (KPIs) for both environmental and economic aspects. The performance indicators will be presented in the next subsections as needed.

5.3.1. Managerial insights from the single grade deterministic model

Our focus from SGDM model is to determine the interval of influence of each unit cost and to analyze its impact on the following KPIs:

- *Collection efficiency (CE)*: the ratio of the collected items to the total demand for refurbished and new items and is used as an ecological indicator.

$$CE = 100 \times \frac{q^c}{d^a + d^r}$$

- *Amount of disassembled part used in manufacturing (UPIN)*: indicates the value of integrating both manufacturing and refurbishing processes by sharing the returns.

$$UPIN = 100 \times \frac{q^u}{q^a}$$

- *Substitution (SUB)*: is an indicator that contributes to the overall flexibility of the production system, enabling manufacturers to respond promptly and effectively to the demand for refurbished products, thereby improving customer satisfaction.

$$SUB = 100 \times \frac{q^s}{d^r}$$

- *Lost sales (LOST)*: a metric used to quantify the amount of unfulfilled customer demand for refurbished products and is used as a customer service indicator.

$$LOST = 100 \times \frac{L^r}{d^r}$$

- *Net Profit Ratio (NPR)* : It indicates the ratio of net profit at a given value of a unit cost ($c\cdot$) to the net profit obtained at the lowest value of this cost. This measures the sensitivity of profit in relation to variations in cost.

$$NPR = 100 \times \frac{Profit(c\cdot)}{Profit(c\cdot = 0)}$$

All parameters were set to their default values, unless indicated otherwise. The variation of each unit cost is made within its logical interval and expressed as a percentage of the manufacturing cost c^a. Each test is carried out for three different values of the price of refurbished items P^r: $P^r = 1.5c^a$, $P^r = 2.25c^a$, $P^r = 3c^a$.

(a) Effect of varying collection cost

Fig. 5 shows the effect of varying collection cost on the four KPIs: CE, UPIN, SUB, and LOST. The collection cost is varied from 0% to 30%

Table 5
Performance of the TRF-FO heuristic for instances with $T = 24$ time periods. CPU CPLEX = 100 s.

Parameter		Case 1: Base case		Case 2: No line sharing		Case 3: Zero changeover time	
		TRF-FO CPU (s)	TRF-FO GAP (%)	TRF-FO CPU (s)	TRF-FO GAP (%)	TRF-FO CPU (s)	TRF-FO GAP (%)
D_{Var}	Low	7.27	0.02	5.00	0.00	5.42	0.10
	High	4.58	0.10	4.57	0.16	4.95	0.14
D_{Val}	Low	2.27	0.72	2.24	0.69	2.26	0.49
	High	4.28	0.04	4.28	0.04	4.30	0.04
SETUP	Low	2.72	0.00	2.74	0.00	3.04	0.00
	High	2.39	0.63	2.38	0.63	2.16	0.65
Changeover time	Low	4.96	0.06	5.05	0.03	–	–
	High	5.31	0.00	5.26	−0.02	–	–
Average		4.22	0.20	3.94	0.19	3.69	0.24

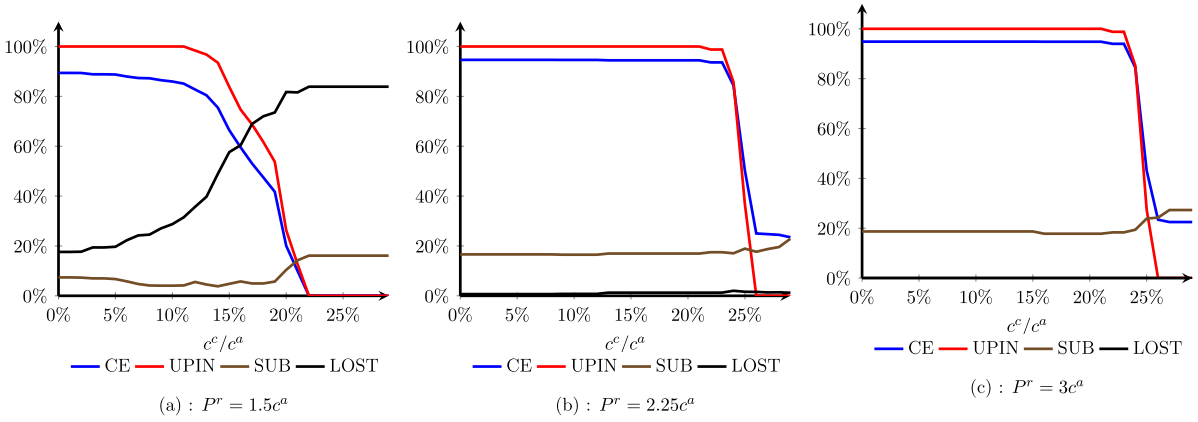


Fig. 5. Effect of varying collection cost on the KPIs.

of the new product assembly cost. Fig. 5(a), (b), and (c) correspond to refurbished items selling price $P^r = 1.5c^a$, $P^r = 2.25c^a$, and $P^r = 3c^a$, respectively.

The results indicate that as the collection cost increases, the collection efficiency decreases together with the amount of used parts (from disassembly) assembled in the new finished product. This is the opposite for substitution which increases as collection cost increases. High collection costs discourage collection and usage of used products. When collection cost becomes too high, it becomes more interesting to satisfy demand for refurbished items through substitution rather than through refurbishing.

We notice that there are three intervals: two stability intervals where collection cost has no effect on the three indicators and the third interval in the middle where major changes take place. By analyzing the three figures together, we notice that when the collection cost c^c reaches a certain percentage of P^r, the collection efficiency (CE) and the amount of disassembled part in the new product drop quickly. This is more remarkable when P^r is high in Figures (b) and (c). With an increase of 3% in the collection cost, collection efficiency decreases by 75%. It becomes more relevant in this case to purchase new components rather than collecting and disassembling used products. Beyond another value of c^c at the end of the middle zone of the figures, it becomes more relevant to either accept lost sales on the refurbished items or go for substitution. Beyond this value, the total collection and refurbishing cost becomes higher than the selling price of the refurbished products (P^r), which would generate negative profits.

We can notice that lost sales decrease considerably when the selling price of refurbished items P^r increases from $1.5c^a$ to $3c^a$. Notice that in Fig. 5 case (a) where P^r is small ($1.5c^a$), collection cost c^c has a higher impact on the variation of lost sales and substitution. The range of variation of the indicators is much wider (Between $c^c = 3\%$ and $c^c = 2.25c^a$). In case (c) of Fig. 5, on the other hand, the range of c^c is from 23 and 27% of c^a. Below or above this range, the system is not sensitive to variation of c^c.

For the decision maker, these experiments indicate the threshold value of c^c from which collection efficiency decreases considerably. However, this efficiency goes down to zero when the selling price P^r is low but stabilizes around 20% when P^r is high. It is interesting to note that this difference in interval length (the middle interval) is useful in the case where c^c becomes a decision variable: The range of c^c for the decision maker is very narrow in case P^r is high.

(b) Effect of varying new component cost

An increase in the new component cost motivates the use of collected items to extract the key component and to use them for refurbishing as shown in Fig. 6. This results in higher collection efficiency and higher amount of used components in the new product and a decrease in substitution. It is worth noting that average lost sales are high with low values of P^r ($P^r = 1.5c^a$) and are rather constant.

In the case of average P^r ($P^r = 2.25c^a$), the lost sales of refurbished items are low and keep decreasing as c^n increases. Finally, for $P^r = 3c^a$, lost sales are always zero. When $P^r = 1.5c^a$, as c^n increases, the gap between cost of new items and price of refurbished items increases, which discourages substitution notice that it stops at a maximum of 60% in this case, compared with 90% and 95% on curves (b) and (c) in Fig. 6. Also, because lost sales are not allowed on the demand for new items, collected items are preferably used in the decomposition and satisfaction of the new items instead of refurbishing them. Which also contributes to the increase in lost sales of refurbished items.

We can also notice the quick increase in collection efficiency with large values of P^r. In Figure (b), for example, an increase of 4% in the price of purchased new parts, results in 80% increase in collection efficiency.

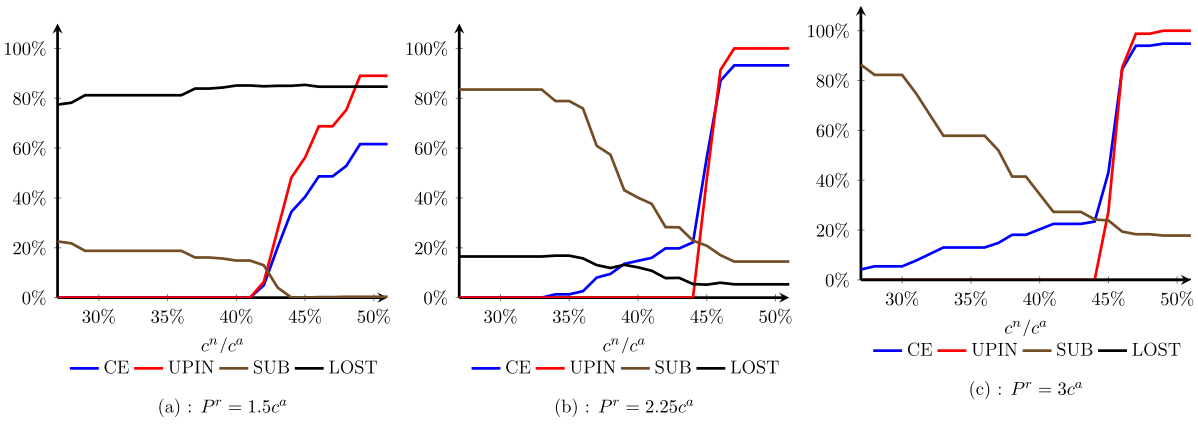


Fig. 6. Effect of varying purchasing cost on KPIs.

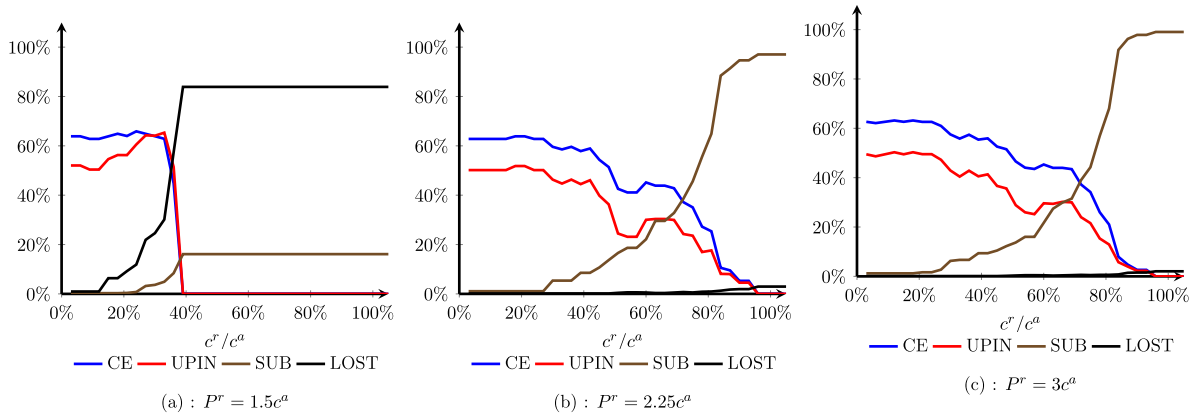


Fig. 7. Effect of varying refurbishing cost on KPIs.

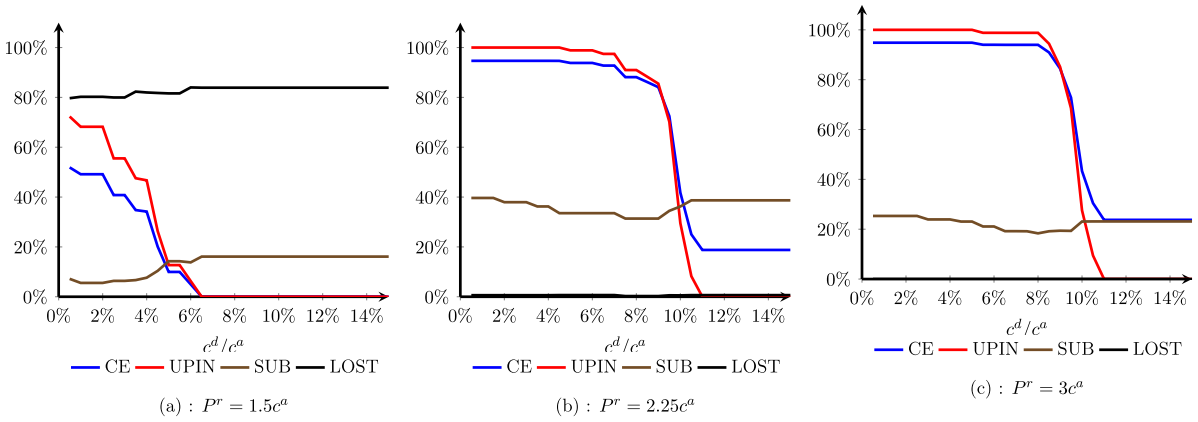


Fig. 8. Effect of varying disassembling cost on KPIs.

(c) Effect of varying refurbishing cost

For curve (a) in Fig. 7, when c^r is low, there are no lost sales. As c^r increases beyond 25% of c^a, it becomes preferable to use collected items in the decomposition and use disassembled part in the new items without allowing substitution (because of the low price P^r of refurbished items). In a very small range, there is a trade-off between substitution and cost of lost sales. The system uses substitution to avoid high costs of lost sales, increasing the amount of decomposed items, until substitution reaches a threshold. Beyond that "reasonable" value of c^r, all demand for d^r is satisfied from substitution and the rest is lost sales (85%). In other words, beyond a certain level of c^r, refurbishing stops and d^r is only satisfied from substitution (Path 1 of Fig. 1). Looking across Fig. 7(a), (b), and (c), when c^r is low, changing P^r has no impact on the decisions: all four indicators start at the same level.

(d) Effect of varying disassembling cost

In Fig. 8, when the disassembling cost c^d increases beyond a certain level, it becomes less interesting economically (to satisfy demand for new items). This drop in level of disassembling, makes collection setup costs unjustifiable, hence it becomes preferable to stop collection completely (and hence refurbishing) and satisfy some of the demand

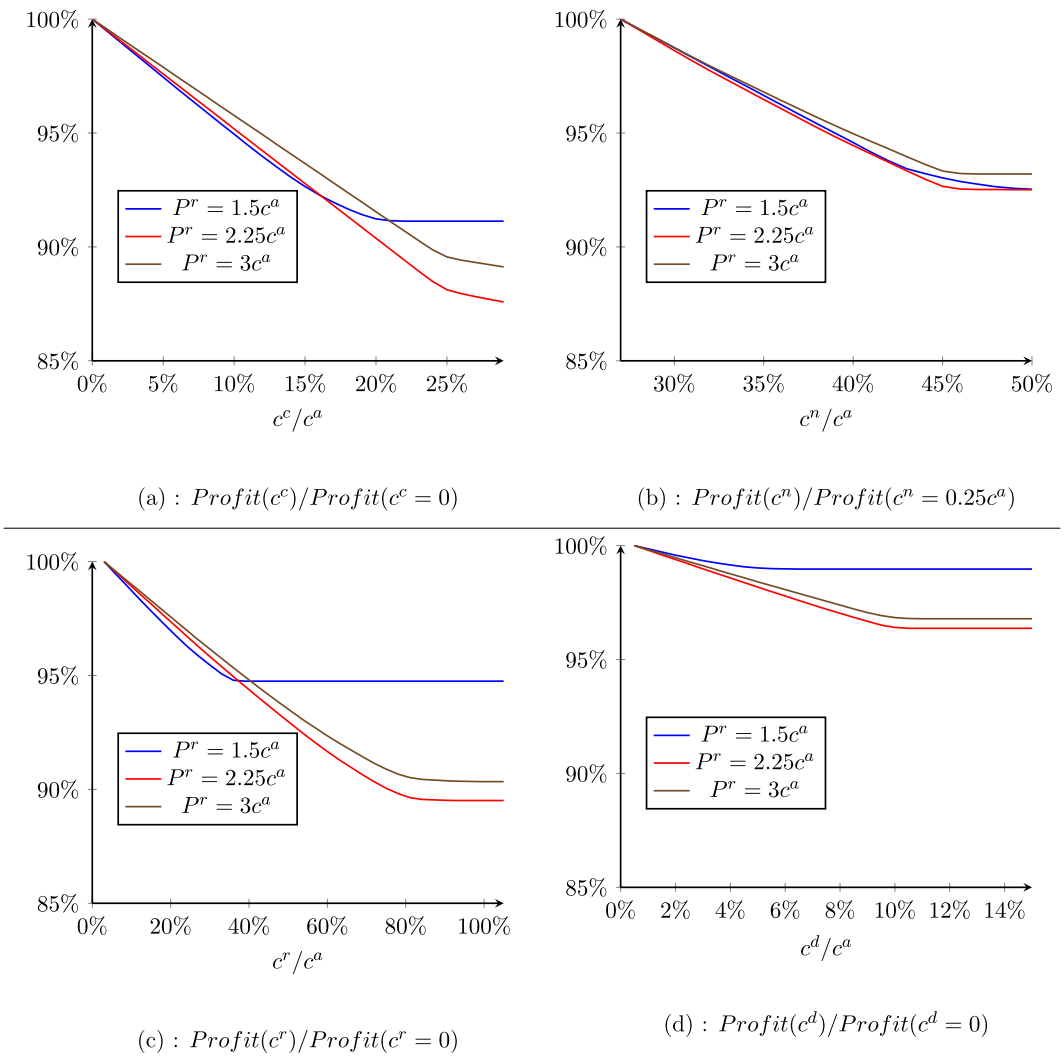


Fig. 9. Profit evolution vs. varying unit costs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

d^r from substitution while accepting an increase in lost sales (as P^r is small).

(e) Analysis of profit evolution:

This paragraph provides a discussion of the variation of total profit as a function of different costs. Fig. 9 shows the percentage decrease in Net Profit Ratio (NPR) described above. The decrease in NPR varies between 3.2% when varying c^d in Fig. 9(d) and 12.2% when varying c^c in Fig. 9(a). We notice that in all cases, the worst decrease in total profit occurs for $p^r = 2.25c^a$ (red curve). The variation of c^n impacts the profit in the same manner regardless of the value of P^r with a fast decrease up to $c^n = 0.45c^a$ and a relative stabilization afterwards. When c^n gets high, we tend to stop purchasing new components. Hence, profit is no longer influenced by c^n variation. Another particular remark in c^n variation is that the profit loss in the case of $P^r = 1.5c^a$ is more important compared to other cost variations in curves (a), (c) and (d) in Fig. 9. Another particularity of c^n is that profit of $P^r = 1.5c^a$ stabilizes after the other profits. The least profit loss happens when varying c^d while $P^r = 1.5c^a$ (Curve (d) in Fig. 9). The profit decreases by less than 1% even at very large values of c^d. This is an important observation showing the small impact of varying decomposition cost on the total profit. Even when $P^r = 2.25c^a$ and $P^r = 3c^a$, the profit loss does not exceed 3.2%.

5.3.2. Managerial insights from stochastic model

This section provides the insights from the stochastic model (SGSM), where demand d^r for refurbished products and the quantity of returned items q^c are uncertain parameters. Their values are identified as a random ratio of demand for new products. A discrete scenario-based approach is adopted to model the uncertainties. As such, the numerical experiments are conducted by considering five realizations ($R_1 < R_2 < R_3 < R_4 < R_5$) for each uncertain parameter. By combining them, a total of 25 scenarios, with equal probabilities, are generated as shown in Table 6. The five realizations are generated uniformly in the interval between the lower value of the ratio (LR) and the higher value of the ratio (HR) as follows:

$$R_i = U(LR+(HR-LR)/5\times(i-1), LR+(HR-LR)/5\times i)\times\mu^a, \quad i = 1,\dots,5,$$

where μ^a is the mean of demand for new products. For example, if $LR = 20\%$ and $HR = 80\%$ then,

$R_1 = U(0.20, 0.32) \times \mu^a$
$R_2 = U(0.32, 0.44) \times \mu^a$
$R_3 = U(0.44, 0.56) \times \mu^a$
$R_4 = U(0.56, 0.68) \times \mu^a$
$R_5 = U(0.68, 0.80) \times \mu^a$

The values of LR and HR characterize the level of uncertainty. We considered 10 levels of uncertainty indexed from 1 to 10 as indicated in Table 7. A level of uncertainty equal to zero corresponds to the deterministic model.

The objective is to analyze the model behavior facing uncertainty and to show how stochastic factors influence the system.

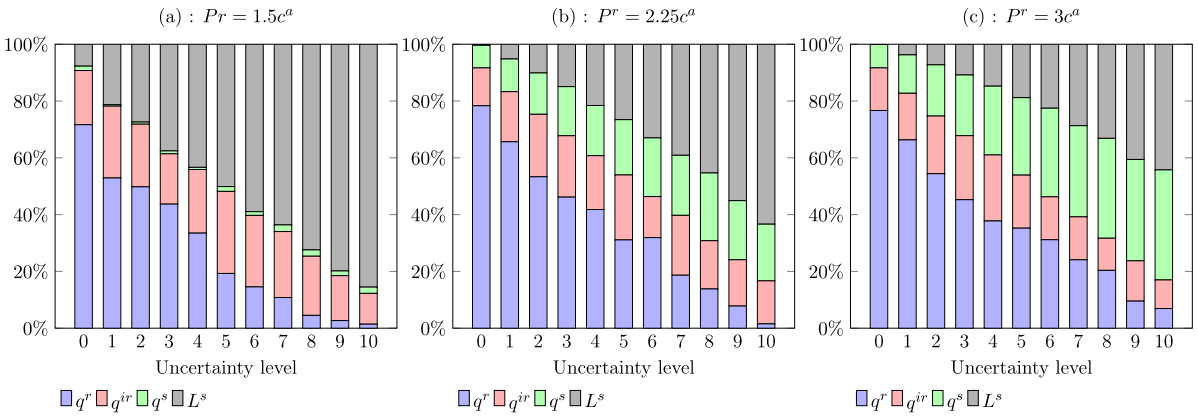


Fig. 10. Rate of refurbishing and substitution in meeting the demand for refurbished products.

Table 6
Scenario generation.

Scenario	Dr	R	Scenario	Dr	R
1	R_1	R_1	14	R_4	R_3
2	R_2	R_1	15	R_5	R_3
3	R_3	R_1	16	R_1	R_4
4	R_4	R_1	17	R_2	R_4
5	R_5	R_1	18	R_3	R_4
6	R_1	R_2	19	R_4	R_4
7	R_2	R_2	20	R_5	R_4
8	R_3	R_2	21	R_1	R_5
9	R_4	R_2	22	R_2	R_5
10	R_5	R_2	23	R_3	R_5
11	R_1	R_3	24	R_4	R_5
12	R_2	R_3	25	R_5	R_5
13	R_3	R_3	/	/	/

Table 7
Levels of uncertainty.

Uncertainty level	0	1	2	3	4	5	6	7	8	9	10
LR	50%	45%	40%	35%	30%	25%	20%	15%	10%	5%	0%
HR	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%	100%

(a) Rate of refurbishing and substitution in meeting demand for refurbished products

Curves (a), (b) and (c) in Fig. 10 present the variation of the quantity refurbished in the dedicated line (q^r), the quantity refurbished in the shared line (q^{ir}), the quantity substituted (q^s) and the lost sales (L^r) as a function of the level of uncertainty for $P^r = 1.5c^a$, $P^r = 2.25c^a$ and $P^r = 3c^a$.

In the deterministic models (uncertainty level = 0), the lost sales level is at minimum value. Its smallest value occurs for the largest value of profit from refurbished items ($P^r = 3c^a$). While q^r decreases with uncertainty increase, the substitution and lost sales level increase. Substitution increases as P^r increases for any uncertainty level.

Figures (a), (b) and (c) show that the total amounts $q^s + L^s$ and $q^r + q^{ir}$ are independent of the selling price P^r for the same level of uncertainty. The upper part ($q^s + L^s$) of the three figures indicates that as profit margin increases, the best option to compensate for lost sales is through substitution instead of refurbishing.

However, $q^s + L^s$ (resp. $q^r + q^{ir}$) increase (resp. decrease) as uncertainty level increases. However, we notice that the amounts produced on the shared line (q^{ir}) are rather independent of the uncertainty level. In the case of high risks (high level of uncertainty), it is better to avoid higher fixed setup costs on the refurbishing line and keep using the shared line. This is different in the case of high profit margins on Figure (c), where the contribution of q^r is higher than in Figures (a) and (b).

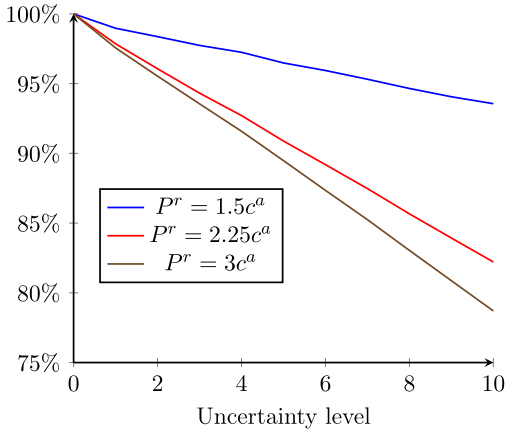


Fig. 11. Effect of uncertainty level on total profit.

Such high price of refurbished items justifies the use of more setup costs on the dedicated refurbishing line.

(b) Effect of uncertainty level on the total profit

Fig. 11 plots the percentage decrease of profit compared to the deterministic case profit (Uncertainty Level = 0) as a function of uncertainty level. This decrease in profit changes considerably between cases with low and high profit margins ($P^r = 1.5c^a$, $P^r = 2.25c^a$, and $P^r = c^a$). Percentage of profit decrease is higher (up to 21% from deterministic case) for $P^r = 3c^a$. For small profit margin of refurbished items, the uncertainty level has much less effect on the total profit decrease (less than 7%). This can be helpful for the decision maker to set an appropriate price level for the refurbished item.

(c) Sharing of costs among different processes

Figs. 12, 13, and 14 provide a discussion of the sharing of costs among the different processes. They are also used to support the discussion given in Section 5.3.2. For a sales price of $P^r = 1.5c^a$, Fig. 12(a), (b) and (c) show the sharing of holding costs, setup costs, and unit costs (respectively) among manufacturing, collection, purchasing and refurbishing as a function of the uncertainty level. Similarly, Figs. 13 and 14 are used for the same analysis for $P^r = 2.25c^a$ and $P^r = 3c^a$. In terms of holding costs, uncertainty level has the highest impact on refurbished and collected Products. This reflects the fact that the uncertainty is on collection and demand for refurbished items. In terms of setup, the impact of uncertainty is higher on setup of refurbishing (decreases as uncertainty increases) and on setup of manufacturing new products (increases with uncertainty level). Finally, in terms of total variable processing costs, the uncertainty mainly impacts refurbishing costs which decrease with increasing uncertainty.

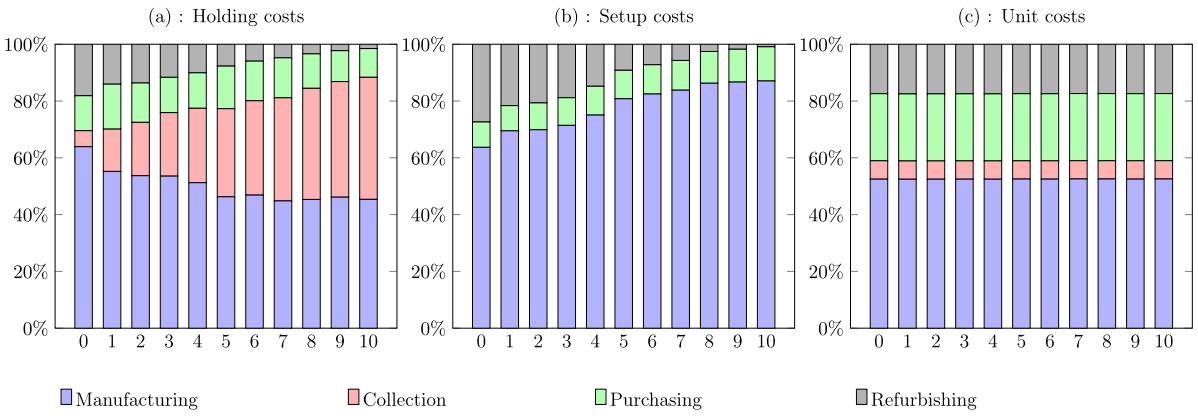


Fig. 12. Contribution of different costs to total cost for $P^r = 1.5c^a$.

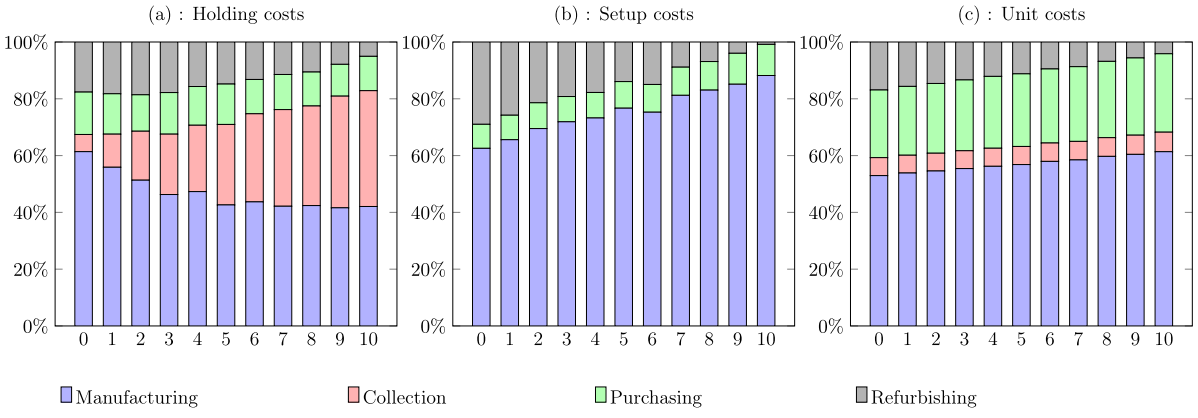


Fig. 13. Contribution of different costs to total cost for $P^r = 2.25c^a$.

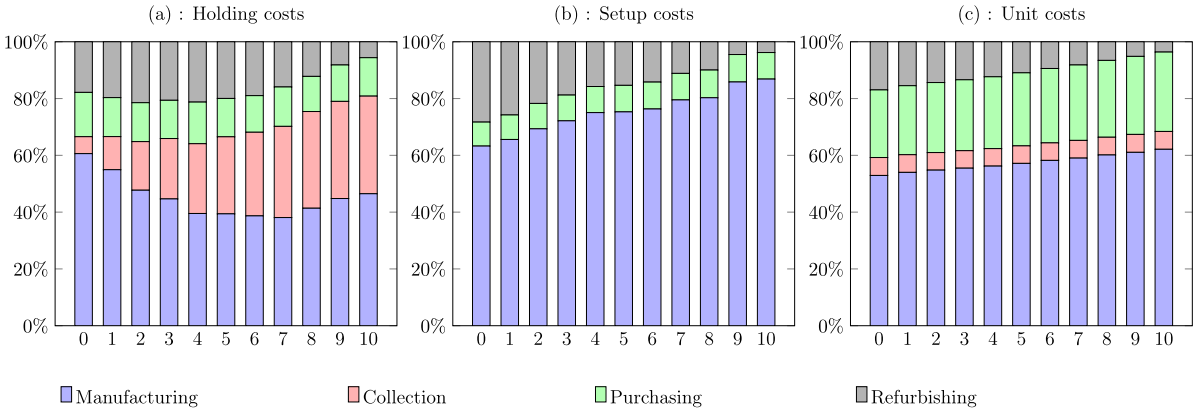


Fig. 14. Contribution of different costs to total cost for $P^r = 3c^a$.

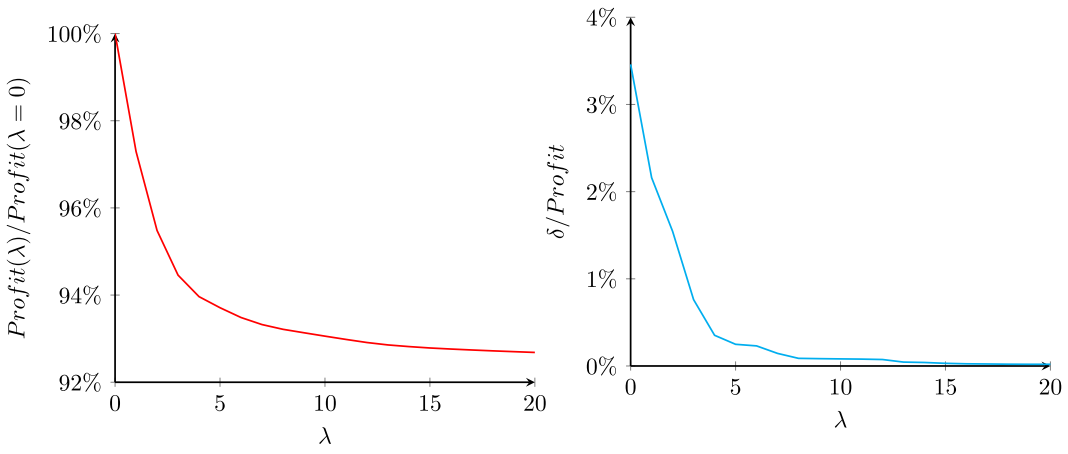
(d) Analysis of the risk averse model

Fig. 15(a) and (b) plot the variation of profit reduction and risk, respectively, as a function of the risk parameter λ. In plot 15(a), the reference value from which reduction percentage is calculated corresponds to the profit at $\lambda = 0$. Preliminary tests suggested that it is sufficient to vary the risk parameter λ in the interval [0, 20] to analyze the trade-off between the decrease in the expected total profit and the reduction in the expected deviation.

These results indicate that to decrease variability from 3.5% to 0 and ensure robustness of the model, we need to sacrifice 7% of profit, which corresponds to the price of robustness. Comparing risk reductions with expected profit reduction, More specifically, we can notice that for $\lambda = 4$, the risk is reduced by more than 3% compared to $\lambda = 0$, while the total profit decreased by 6%.

5.3.3. Managerial insights from the multi-grade multi-quality level model

For the Multi-Grade Deterministic Problem, there are three levels of quality, $i = 1, 2, 3$, corresponding to low, medium, and high quality, respectively. These levels can be viewed also as basic, standard, and premium quality levels. The parameter default values of the medium quality ($i = 2$) are generated exactly as indicated in Section 5.1. The holding cost $h_1^a = 0.5$, $h_2^a = 1$, $h_3^a = 1.5$. The manufacturing cost for each grade quality i is $c_i^a = h_i^a/0.15$. The other holding and unit processing costs are set as indicated in Section 5.1. The upgrading costs are a



(a) : Expected profit

(b) : Variability of profit

Fig. 15. Profit and variability evolution vs. varying λ.

Table 8
Description of nodes in Sankey diagrams.

Nodes	Description
Purchase	Quantities purchased
QC	Quantities collected
QR	Quantities refurbished in the dedicated line
Qir	Quantities refurbished in the shared line
LR	Quantities lost of demand for refurbished items
QA	Quantities manufactured
DA	Demand for new items
DR	Demand for refurbished items
QD	Quantities disassembled

percentage of manufacturing costs as follows: $c^{up}_{(1,2)} = 0.2c^a_2$, $c^{up}_{(1,3)} = 0.4c^a_3$, $c^{up}_{(2,3)} = 0.3c^a_3$. All other parameters which are independent of the subscript i are kept as in Section 5.1. The flow quantities correspond to the cumulative amounts over the whole planning horizon.

Sankey diagrams are used to analyze and to represent the magnitude of flows between different nodes in the system through various processing stages to the final products. The description of nodes is given in Table 8.

The results are presented in Figs. 16, 17, 18, and 19.

Fig. 17 shows the flows when the gap between quality levels is low or high in subfigures (a) and (b), respectively. The gap is reflected in the difference between the holding costs of different levels. For low-quality gap, $h^a_1 = 0.95$, $h^a_2 = 1$, $h^a_3 = 1.05$. For high quality gap, $h^a_1 = 0.25$, $h^a_2 = 1$, $h^a_3 = 1.75$.

Comparing Fig. 16(a) and (b), we can notice that downgrading is proportional to the quality gaps. When quality gap increases, lost sales occur mostly on the lowest quality product in this case, sharing of the production line between refurbishing and manufacturing increases for higher quality products, and substitution decreases.

The second experiment concerns the consideration of finite vs. infinite storage capacity of returns. In the former case, the storage capacity is limited to two periods of returns. The results are shown on Fig. 17(a) and (b), respectively. When inventory capacity is limited, lost sales are higher while disassembly process is not activated at all. We can also notice that the shared line is utilized at higher volumes when the capacity is infinite. In this same situation, substitution for low-quality products has increased.

In Fig. 18(a) and (b) we discuss the impact of the amount of returns (as a percentage of new items' demand). It is easy to observe the increase in purchased new parts and substitution where returns rate is low in Figure (a). We have on the other hand an increase in the amount of disassembled returns when their rate is high as shown on Figure (b). In this case, refurbishing also increases both on the dedicated and the shared line. However, upgrading from low-quality to high quality products decreases considerably in this situation.

Finally, Fig. 19 shows the impact of upgrading cost on the flows. The most remarkable result here is the fact that higher upgrading costs result (Figure (b)) result in higher amounts of disassembled items, higher substitution of low-quality items and lower substitution for higher quality items.

6. Conclusions and future work

We studied the production planning problem in a facility that receives returned products. We have proposed a profit maximization model for dynamic lot sizing considering substitution and multiple usage of returns for refurbishing at different levels of quality or for disassembly to extract key parts that will be used in the manufacturing process. Discussion started with a single grade (single level of quality) and was extended to multi grade case. This model allowed to study interactions between different types of returns and decisions. The impact of uncertainty on the different interactions was analyzed through a robust two-stage stochastic program with uncertainty on demands and returns. The models were implemented as mixed integer linear programming formulations. These optimization problems are NP-hard. Therefore, we proposed an efficient heuristic based on relax-and-fix and fix-and-optimize approaches. The heuristic give solutions very close to optimum in few seconds on problems with reasonable lengths of the planning horizon.

Several managerial insights were derived through extensive numerical experiments. The main findings are as follows. From an economical point of view, the profit margin of the refurbished items is the main determinant of the total profit and when such margin in high, the total profit becomes more sensitive to different cost variations. With respect to the collection efficiency (environment-related performance indicator), it becomes very sensitive to collection cost and much less sensitive to refurbishing cost. This was more remarkable with high profit margins on the refurbished items. A third finding was related to the two-stage stochastic model. As uncertainty level increases, risk of stock-out increases. The best option chosen by the model to compensate for these stock-outs is substitution instead of more refurbishing. Based

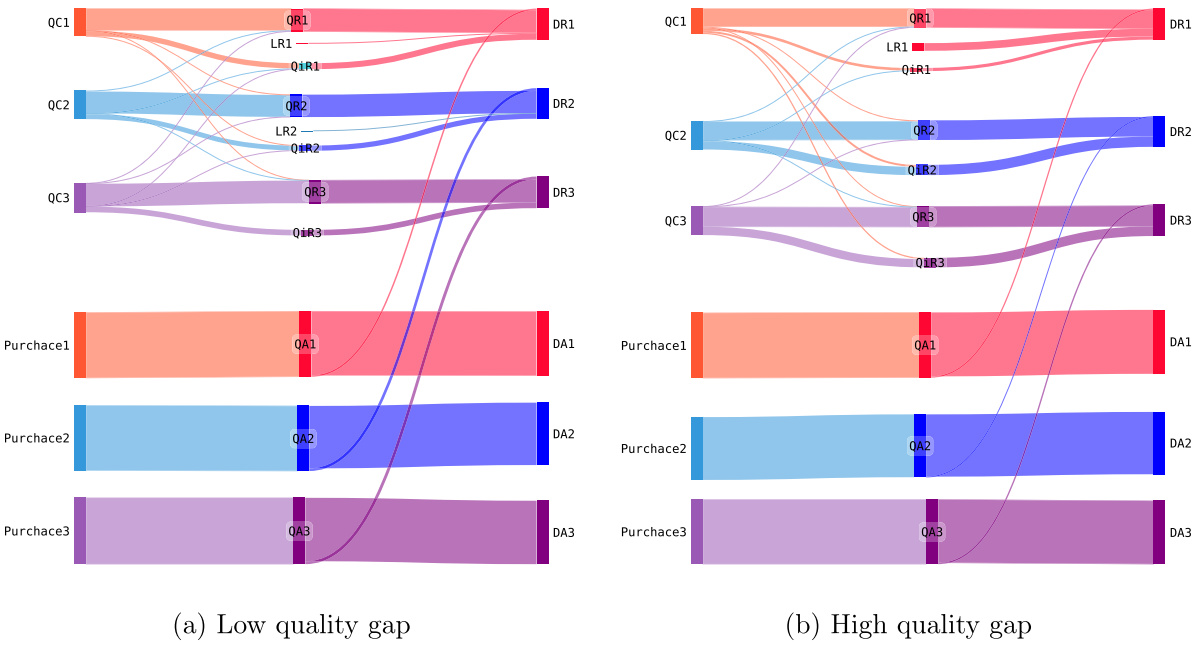


Fig. 16. Flows in case of (a) low vs. (b) high quality gaps.

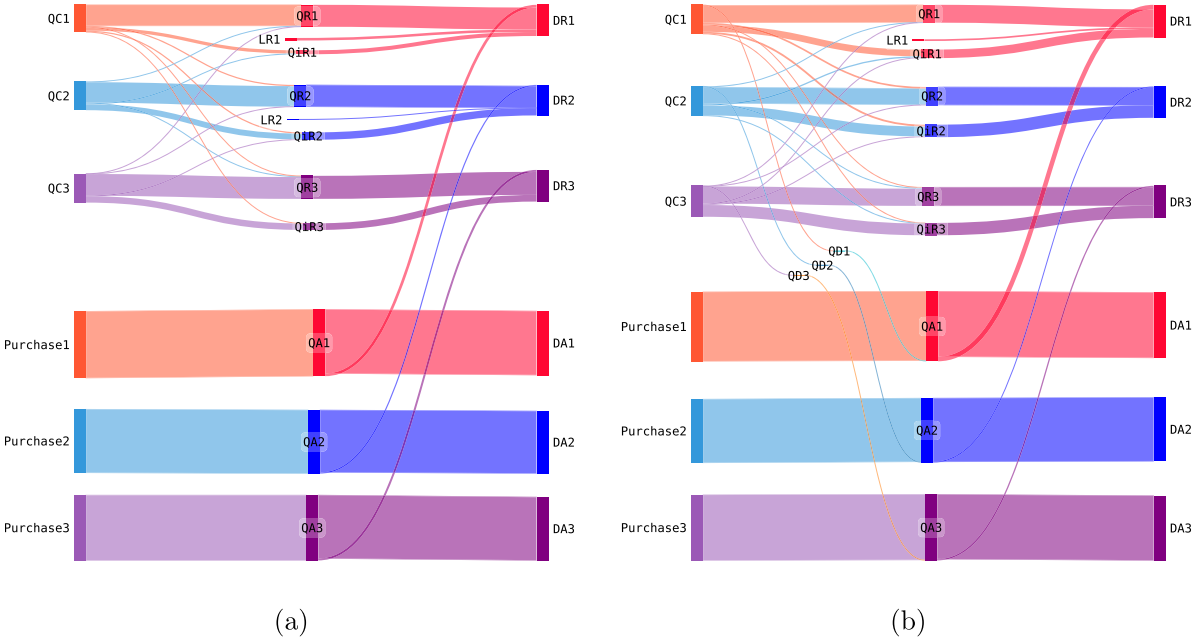


Fig. 17. Returns storage capacity: (a) finite vs. (b) infinite.

on the numerical experiments, the cost of robustness was 7% of the deterministic profit. That is, to ensure robustness of the model, we need to sacrifice 7% of the total profit. In the multi grade case, we have noticed that disassembly is considered as a preferred option mostly when upgrading cost (from low-quality to high quality products) is high. In this same situation, substitution occurs mainly on low-quality products.

An interesting extension of the present work is to evaluate how different types of warranties on new and refurbished products affect lot sizing decisions. It will also be interesting to study the risk of obsolescence of refurbished items especially in the electronic products. Beyond a certain time limit, electronic products cannot be refurbished any more. In addition to the above extensions, there is clearly a need to study the link between product life cycle and refurbishing decisions. It will be useful to extend inventory management policies for remanufacturing (ex. Bayındır et al., 2007) to integrate obsolescence (See also Pierskalla, 1969; Van Jaarsveld and Dekker, 2011). In our models, the environmental impact of integration and substitution was measured through the "collection efficiency" indicator (Section 5.3).

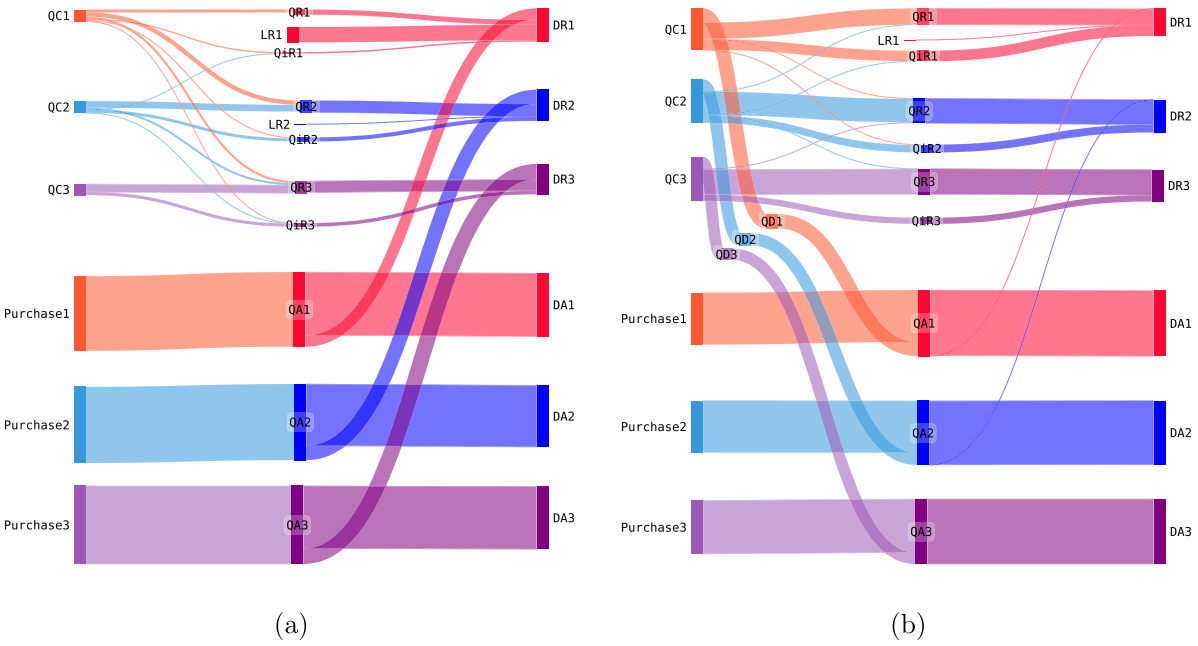


Fig. 18. Percentage of returns: (a) low vs. (b) high.

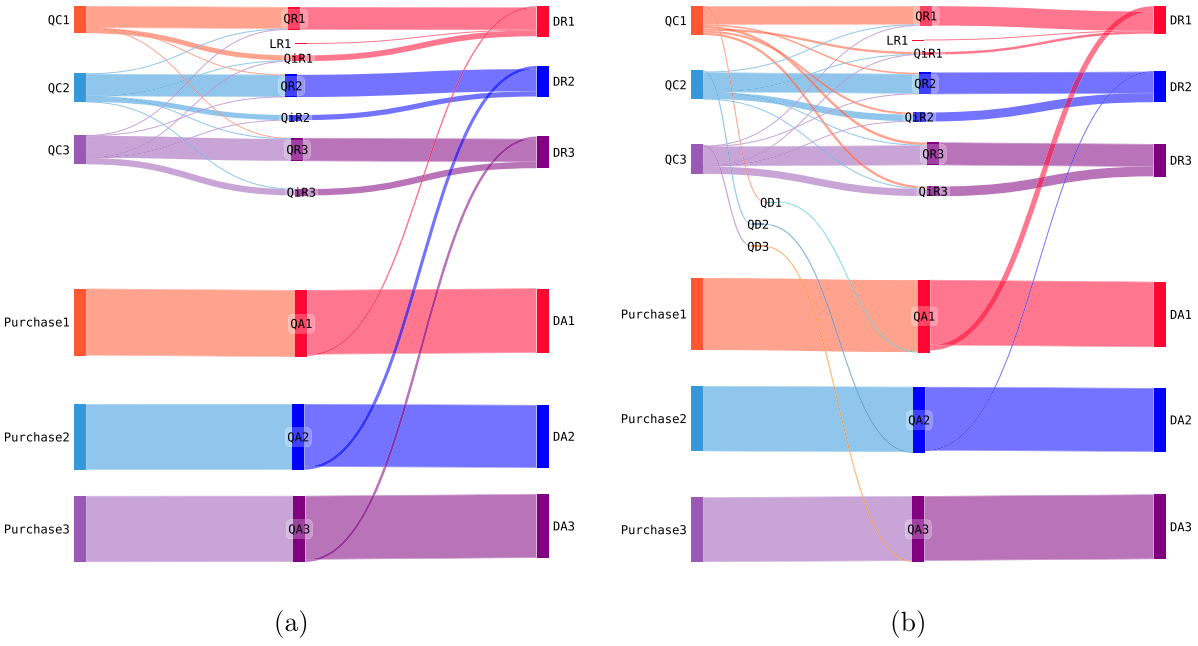


Fig. 19. Upgrade cost: (a) low vs. (b) high.

An important extension of our work could be the development of bi-objective models considering an ecological objective (such as the one mentioned previously) together with an economic indicator (cost or profit).

CRediT authorship contribution statement

Y. Boutarfa: Writing – original draft, Methodology, Formal analysis, Conceptualization. **A. Senoussi:** Writing – original draft, Methodology, Formal analysis, Conceptualization. **N. Brahimi:** Writing – original draft, Methodology, Formal analysis, Conceptualization. **T. Aouam:** Writing – original draft, Methodology, Formal analysis, Conceptualization.

Data availability

Data will be made available on request.

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